The Evolution of the Welfare State

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University of Chile
Department of Economics

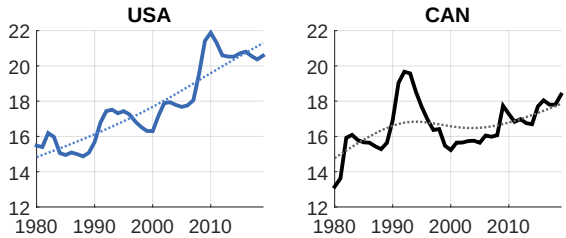
October 15, 2025

Social benefits evolve differently across the world

- Social benefits, share of GDP (e.g. health, family, unemployment).

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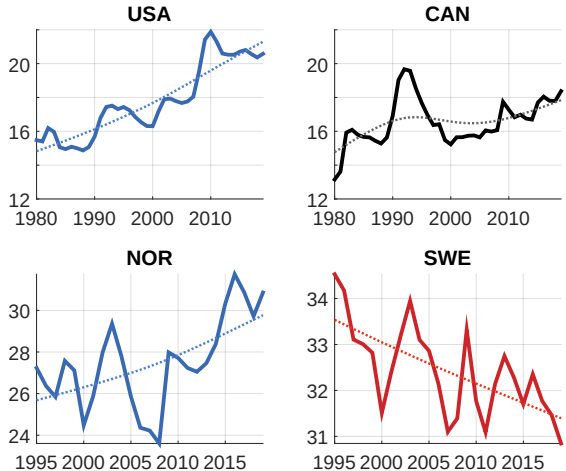
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Source: OECD (2023).

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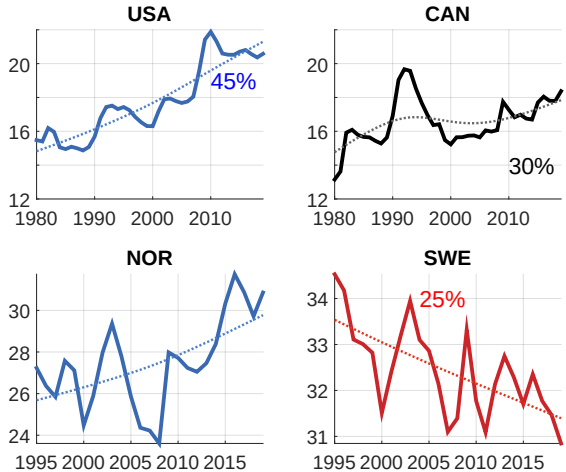
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Net soc. ben.

What explains the evolution of the Welfare State?

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- **This Paper**

Wealth Inequality → Welfare State (+)
Meltzer and Richard (1981)

Social Mobility → Welfare State (-)
Benabou and Ok (2001)

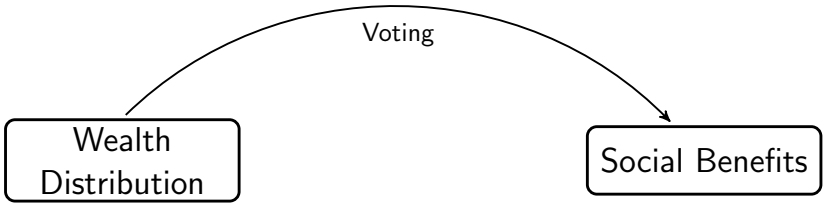
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Wealth Distribution

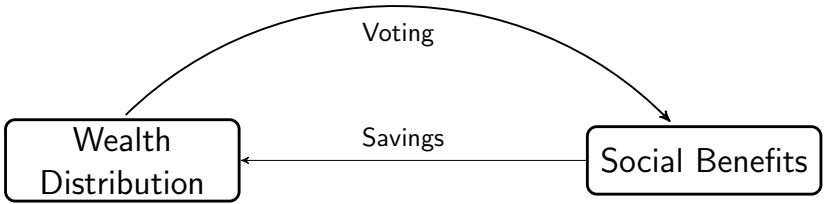
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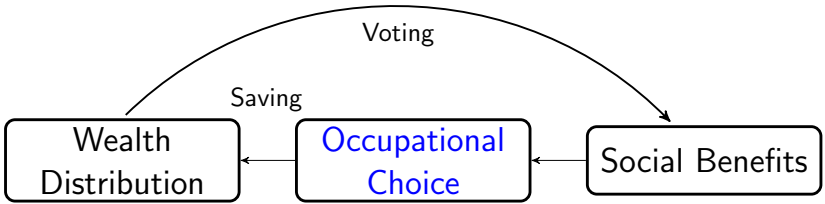
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1. **Inequality-Policy Link**

What explains the evolution of the Welfare State?

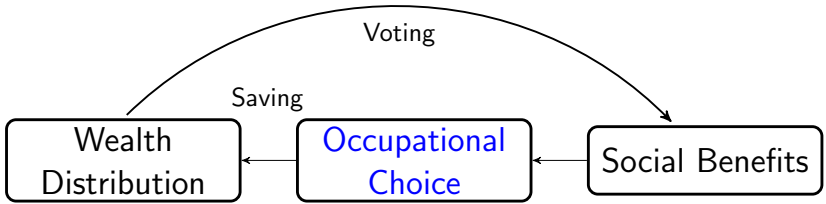
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1. **Inequality-Policy Link**
   ↑
2. **Anticipatory Voting**

What explains the evolution of the Welfare State?

- **This Paper**



1. **Inequality-Policy Link**
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- The **inequality-policy** link predicts a large fraction of countries

Paper's organization and preview of results

1. **Theoretical** analysis of the primitives that drive different trends of the welfare state

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   - Evolution of the Welfare State depends on Average Wealth and Inequality (1st and 2nd moments of the distribution).

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1. **Theoretical** analysis of the primitives that drive different trends of the welfare state
   - Evolution of the Welfare State depends on Average Wealth and Inequality (1st and 2nd moments of the distribution).

2. **Quantitative analysis** in 24 countries to test the theory
   - Model **predicts** $\frac{3}{4}$ of the countries

Contributions to the Literature

1. Theory that explains the differences in the evolution of the Welfare State
   Alesina and Rodrik (1994); Hassler et al. (2003); Alesina and Angeletos (2005)
   - Key role of endogenous class of aspirational voters
     Piketty (1995); Benabou and Ok (2001); Alesina and La Ferrara (2005)

2. Tractable model with heterogeneous agents, occupational choice, and politics
   Krusell et al. (1996); Krusell and Rios-Rull (1996, 1999); Nuño and Moll (2018); Itskhoki and Moll (2019)
   - Theoretical results for transition dynamics

Plan

1. The Model
2. Political Equilibrium
   - 2.1. Individual Preferences
   - 2.2. Probabilistic Voting
   - 2.3. Equilibrium Size of the Welfare State
3. The Evolution of the Welfare State
4. Quantitative Exercise
5. Extensions
6. Conclusions

The Model

The Model

- Continuum of agents heterogeneous in wealth $a_t \sim \Gamma_t(a)$

$$\max_{\{c_t\}_{t=0}^{+\infty}} \left\{ \int_0^{\infty} e^{-\rho t} \log(c_t) dt \right\}$$

$$s.t. \qquad \dot{a}_t = (r - \tau_t) a_t - c_t + \begin{cases} w_t \ell + \boldsymbol{T_t} & \text{if worker} \\ \Pi_t & \text{if entrepreneur} \end{cases}$$

$$a_t \geq \underline{a}$$

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  - Transfer rate: $b_t \geq -\underline{b}$ (social benefits, % of GDP)

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- Transfers to workers: $T_t = \boldsymbol{b_t} \cdot Y_t$
  - Transfer rate: $b_t \geq -\underline{b}$ (social benefits, % of GDP)
- Balanced budget: $\tau_t \cdot A_t = T_t \cdot \boldsymbol{(1 - e_t)}$ ($\boldsymbol{e_t}$: % of entrepreneurs)

Spending Trade-off: Evidence

Two Technologies

1. **Entrepreneurs produce physical capital (K):**

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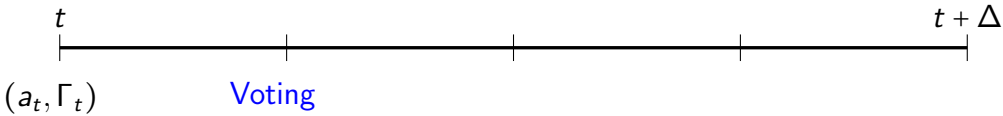
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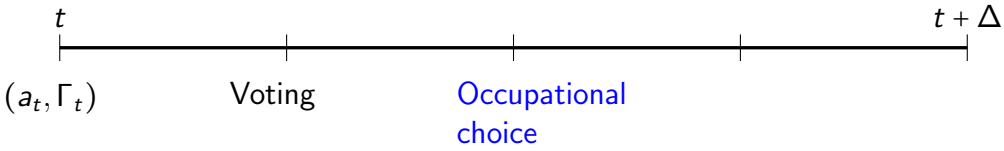
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   Individual profits: $\Pi_t = p_t R - rI$

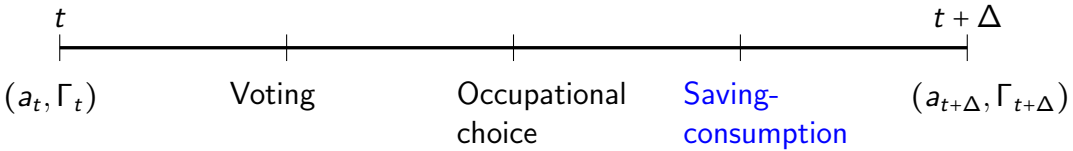
Timing of individual decisions



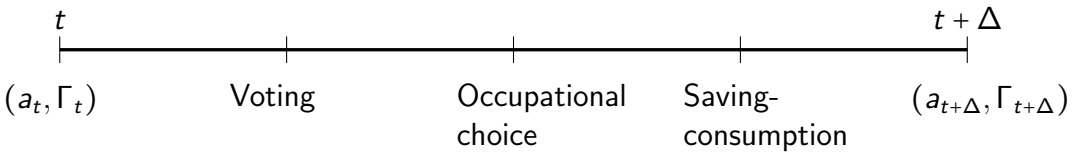
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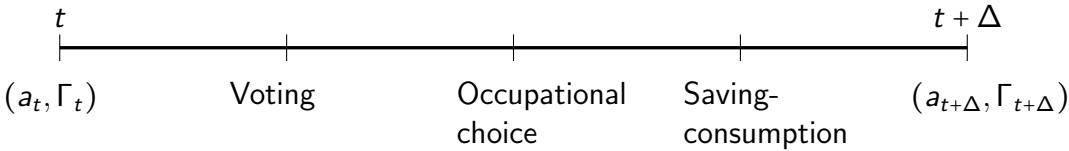


Timing of individual decisions



- **Assumption** Agents expect social benefits to remain stable ($\{b_s\}_{s=t}^{+\infty} = b_t$)
  Benabou and Ok (2001); Alesina and La Ferrara (2005)
  - Maximize lifetime-utility $\Leftrightarrow$ Maximize disposable income

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  - Maximize lifetime-utility $\Leftrightarrow$ Maximize disposable income
- *Alternative:* fully-rational equilibrium (numerical)
  Krusell and Rios-Rull (1996, 1999); Quadrini and Rios-Rull (2023)

Occupational Choice

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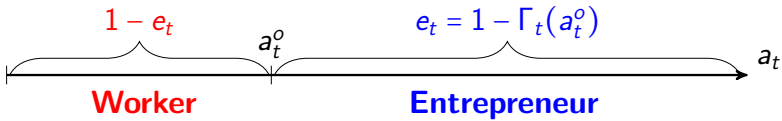
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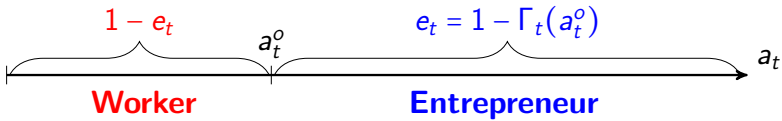
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Occupational Choice



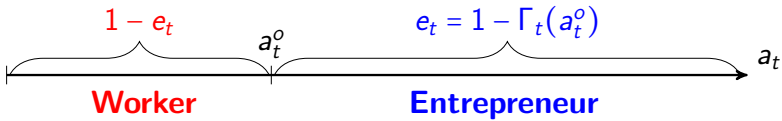
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2. Maximum Sustainable transfer rate: $\overline{b}(\Gamma_t)$

The Model: Main features

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- **Endogenous size of the welfare state (b_t)** (chosen through voting):
  - If $b_t > 0$: workers receive social benefits (pro-worker).
  - If $b_t < 0$: entrepreneurs receive net transfers (pro-business).

Political Equilibrium

Political Equilibrium: Roadmap

1. Individual Preferences

2. Probabilistic Voting (Persson and Tabellini, 2000)

3. Equilibrium Size of the Welfare State

Individual Preferences

Individual preferred transfer rate: $b(a;\Gamma_t)$

- Agents observe a and Γ_t, and maximize disposable income at t:

$$b(a;\Gamma_t) = argmax_{b\in[\underline{b},\overline{b}(\Gamma_t)]}\ y_t(a,b;\Gamma_t) = \begin{cases} y_t^W & \text{if } a < a^o(b,\Gamma_t) \\ y_t^E & \text{if } a \geq a^o(b,\Gamma_t) \end{cases}$$

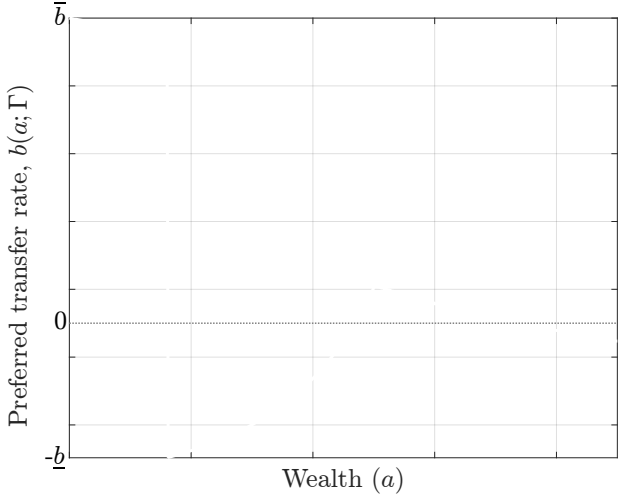
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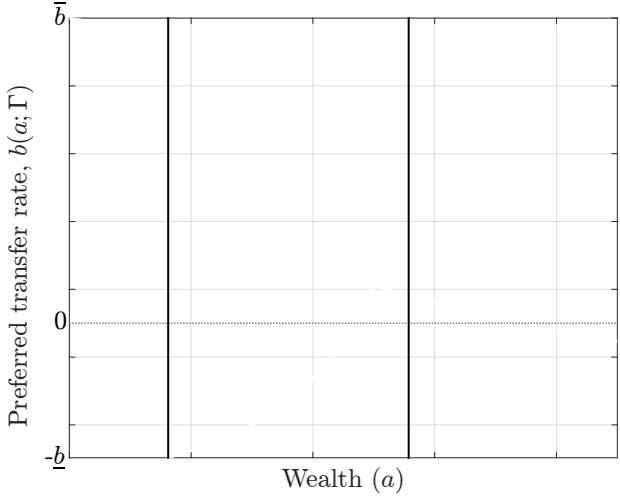
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- Agents anticipate occupational mobility prospects at t

Individual preferred transfer rate

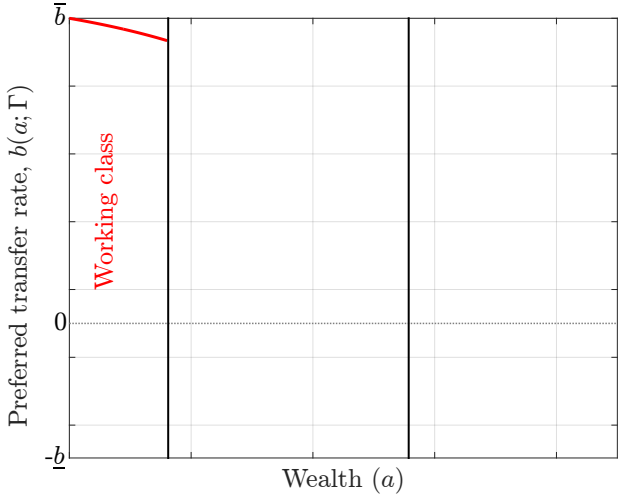


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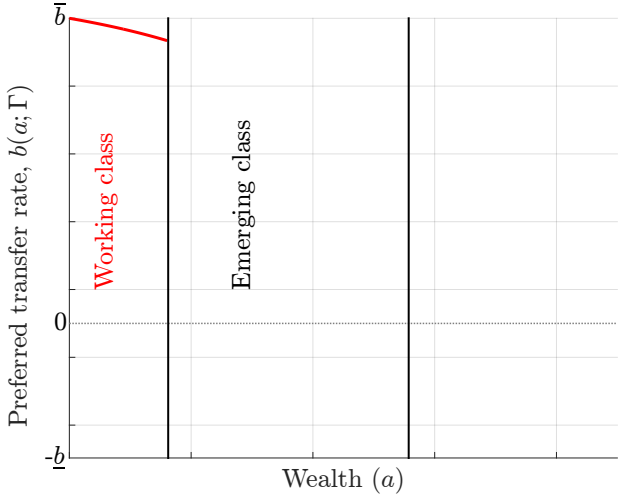
Individual preferred transfer rate

- **Working Class:** high social benefits



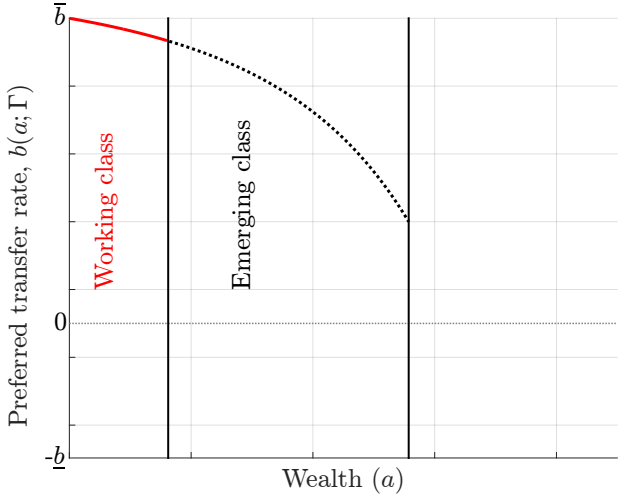
Individual preferred transfer rate

- **Emerging Class:** either Workers or Entrepreneurs



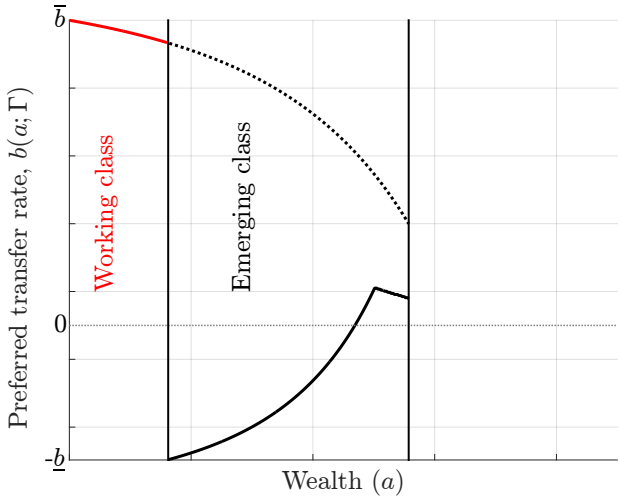
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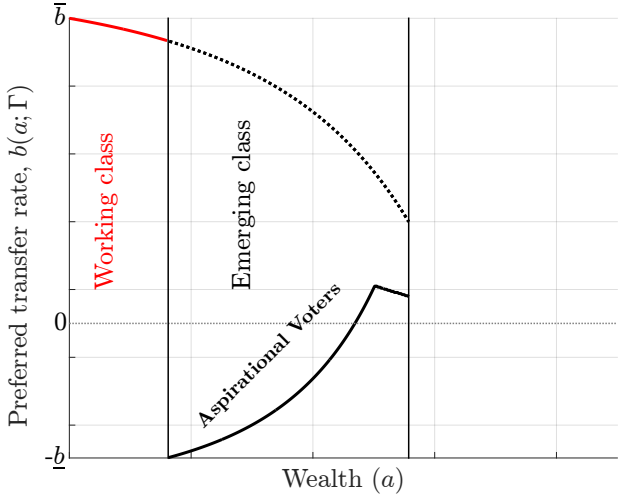
Individual preferred transfer rate

- **Emerging Class:** pro-business policy



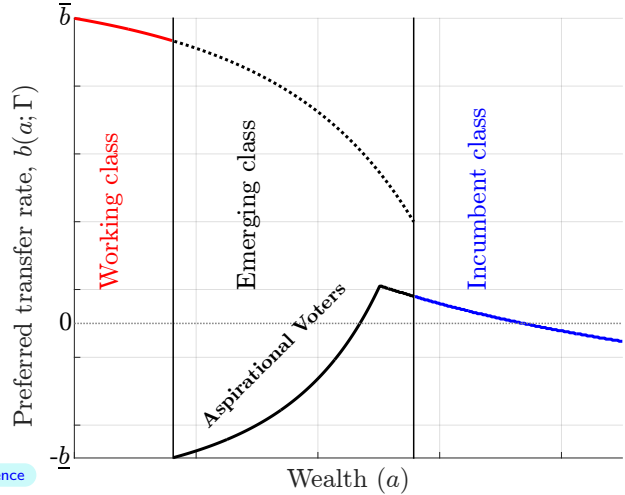
Individual preferred transfer rate

- **Emerging Class:** pro-business policy



Individual preferred transfer rate

- **Incumbent Class:** less pro-business policy



Income function Evidence

Probabilistic Voting

Electoral competition under uncertainty

- Two parties choose b_t^1 and b_t^2 to maximize expected share of votes

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- Voters indexed by (a, p)

  p: idiosyncratic political preference (Uniform, (ϕ^W, ϕ^E))
- Symmetric Nash equilibrium:

$$b_t = argmax_b \left\{ \int_{a<a_t^o(b)} y(a,b) d\Gamma_t(a) + \underbrace{\frac{\phi^E}{\phi^W}}_{\equiv \phi} \int_{a \geq a_t^o(b)} y(a,b) d\Gamma_t(a) \right\}$$

Equilibrium Size of the Welfare State

- Maximize weighted income ($\phi \geq 1$ Political weight):

$$\max_{b} \left\{ w_t \ell \cdot (1 - e_t) + \phi \Pi_t \cdot e_t \right\}$$

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- Maximize weighted income ($\phi \geq 1$ Political weight):

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- Equilibrium policy b_t:

$$\boxed{\mathbf{1} - \Gamma_t(a^o(b_t, \Gamma_t)) = e^* \quad (PE)}$$

  - $e^* = \Psi(Z, r, \alpha, R, I, \ell, \phi) \in (0, \alpha)$

Forward looking gov. | PE: 2-D diagram

Equilibrium Definition

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$$s_t(a) = \theta_t \cdot y_t(a) \qquad (HJB)$$

$$d_t \Gamma_t(a) = H(\Gamma_t, s_t, a_t^o) \qquad (KFE)$$

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$$a_t^o = \max\{\hat{a}_t, \tilde{a}_t\} \qquad (OC)$$

$$e^* = 1 - \Gamma_t(a_t^o) \qquad (PE)$$

$$\tau_t \cdot A_t = T_t \cdot (1 - e^*) \qquad (BB)$$

Stationary Equilibrium

Stationary Equilibrium

1. Unique stationary tax rate: $\boldsymbol{\tau^* = r - \rho}$ $(\theta(\tau^*) = 0)$

2. Set of stationary distributions: $\boldsymbol{\Gamma^*}$

SS details

The Evolution of the Welfare State

The Evolution of the Welfare State

- Six possible patterns for the joint dynamics of (b_t, τ_t, A_t) details

The Evolution of the Welfare State

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Given Γ_0 and $r - \rho$:

1. Γ converges to one Γ^*

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Given Γ_0 and $r - \rho$:

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In all cases, the model converges to a stationary size of the welfare state, b^*

The Evolution of the Welfare State

- Six possible patterns for the joint dynamics of (b_t, τ_t, A_t) details

Main takeaway

1. If $\tau(\Gamma_0) < r - \rho \Rightarrow$ b increasing over time
2. If $\tau(\Gamma_0) > r - \rho \Rightarrow$ b decreasing over time

The Evolution of the Welfare State

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Main takeaway

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Question Which properties of Γ_0 give rise to each case?

Question Which properties of Γ_0 imply that $\uparrow b$ or $\downarrow b$?

- Problem Characterizing distributions is analytically cumbersome

Question Which properties of Γ_0 imply that $\uparrow b$ or $\downarrow b$?

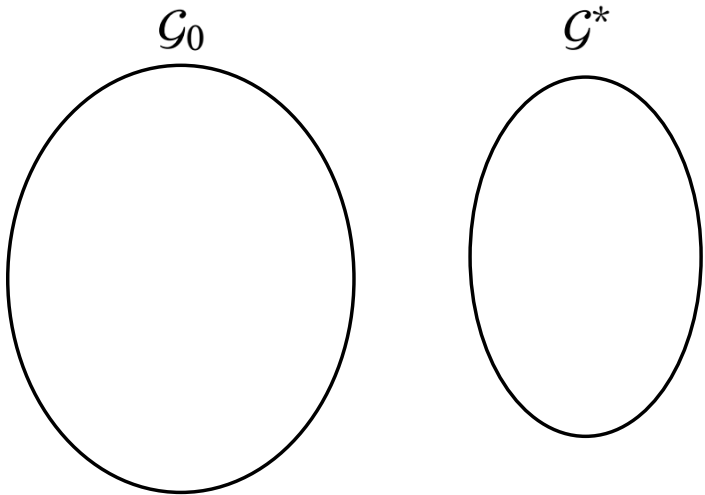
- Problem Characterizing distributions is analytically cumbersome
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Question Which properties of Γ_0 imply that $\uparrow b$ or $\downarrow b$?

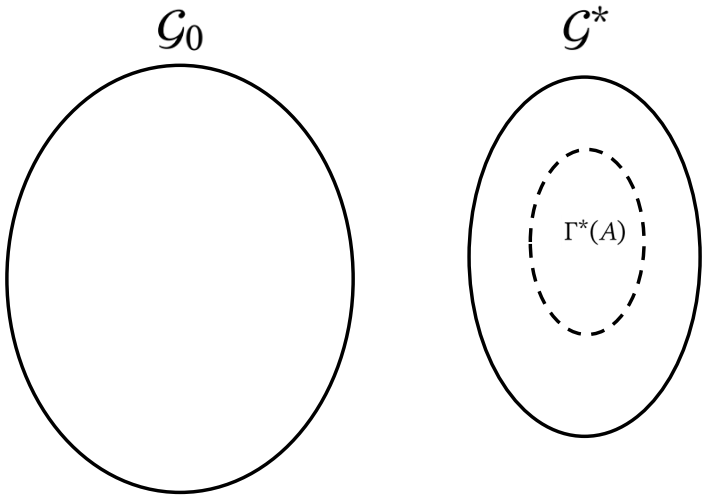
- Problem Characterizing distributions is analytically cumbersome
- **Solution** Construct Γ_0 perturbing stationary distributions Γ^*

  Apply an MPS on Γ^* to obtain Γ_0 (MIT shock)
  MPS around the mean (Rothschild and Stiglitz, 1971)

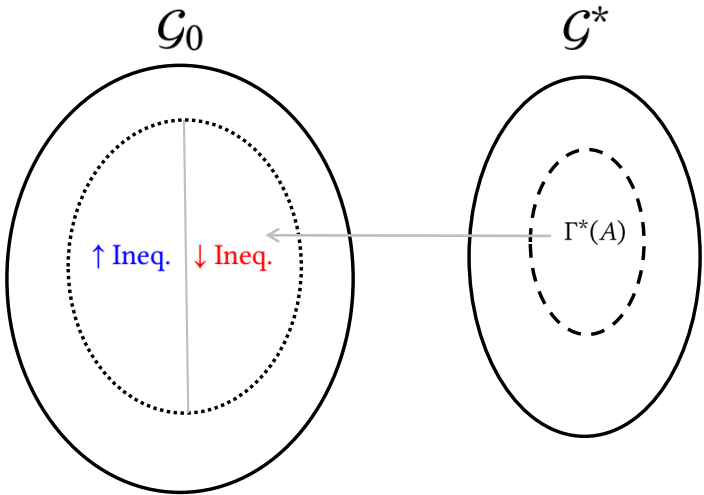
The MPS approach



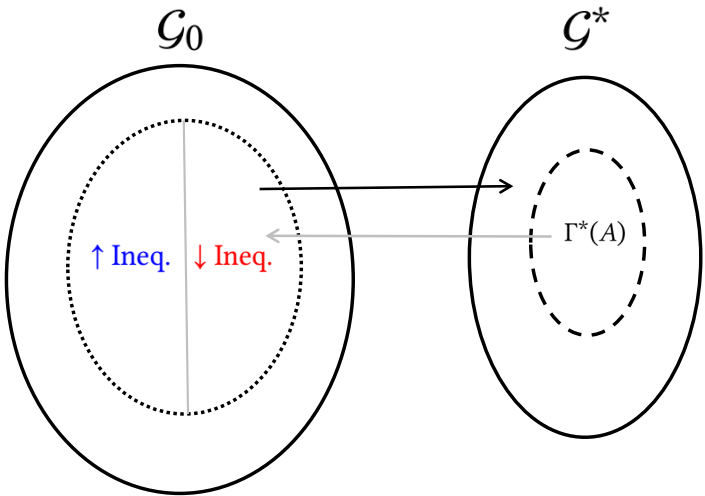
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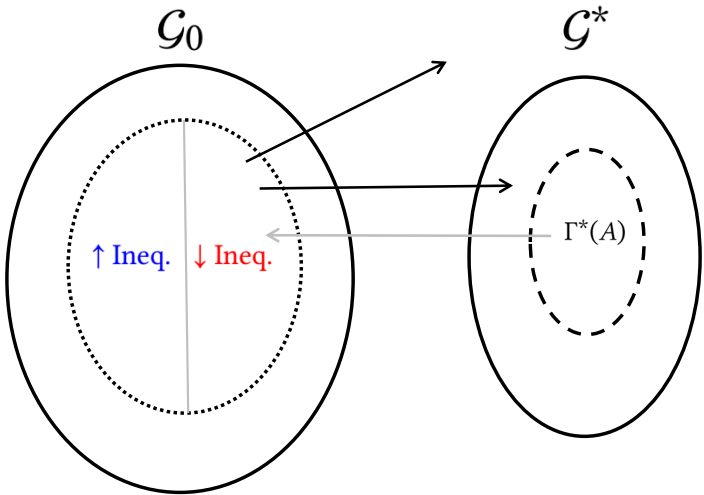
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The MPS approach



Main Theoretical Results

1. Size of Welfare State depends on the mass of **Aspirational** voters

Main Theoretical Results

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2. Evolution of Welfare State depends on Wealth & Inequality

The Evolution of the Welfare State: Wealthy Countries

↑ **Inequality**: USA (1970-2019)

Increasing social benefits

The Evolution of the Welfare State: Wealthy Countries

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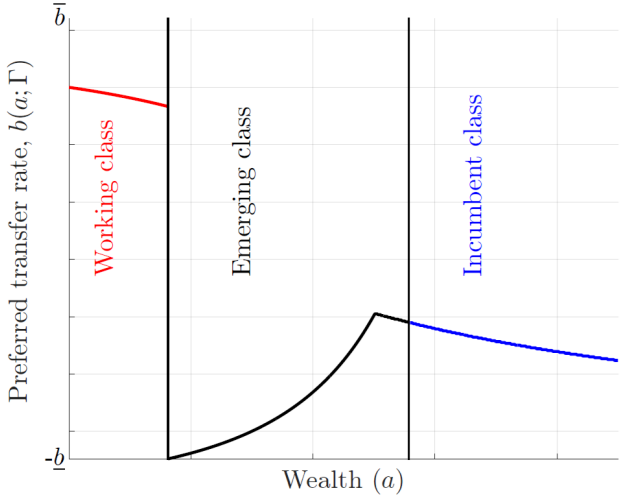
Increasing social benefits

↓ **Inequality**: Sweden (1995-2019)

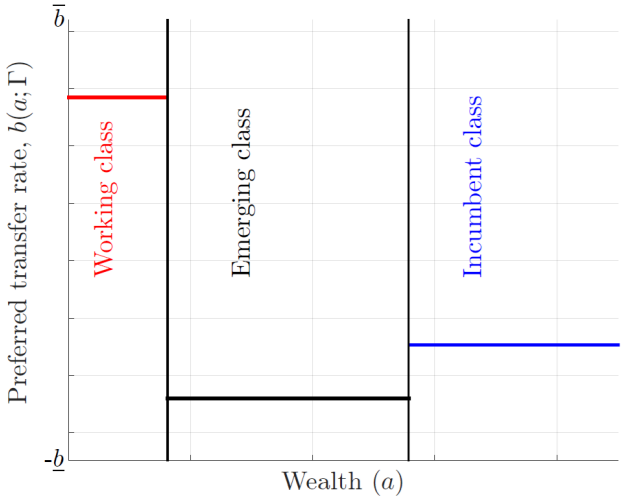
Decreasing social benefits

The American Experience (1970-2019)

American Experience: Intuition

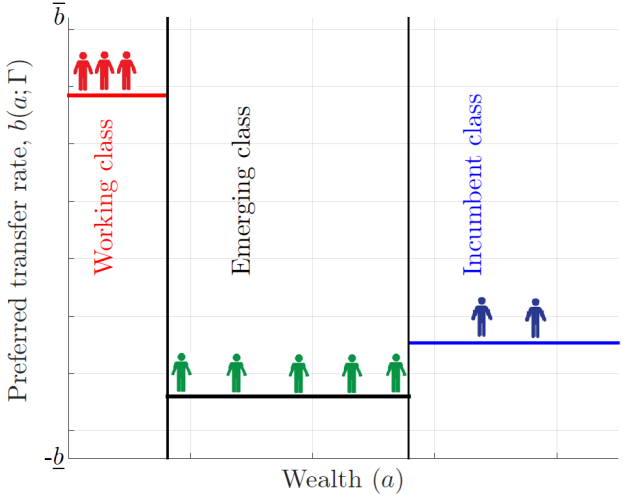


American Experience: Intuition



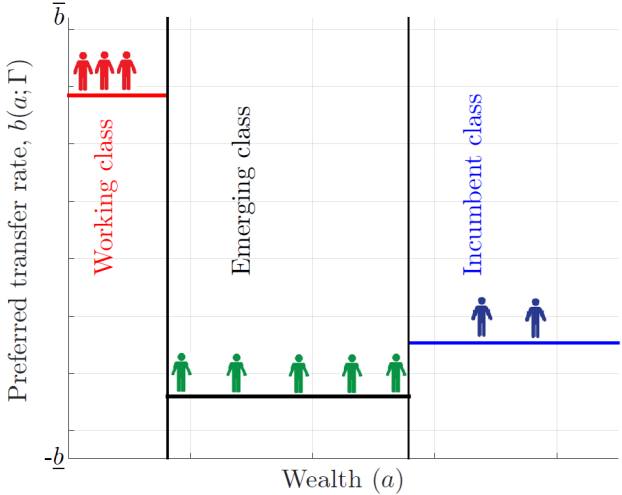
American Experience: Intuition

- $t = 0$: Wealthy and Unequal $\Rightarrow$ Many Aspirational Voters (AV)



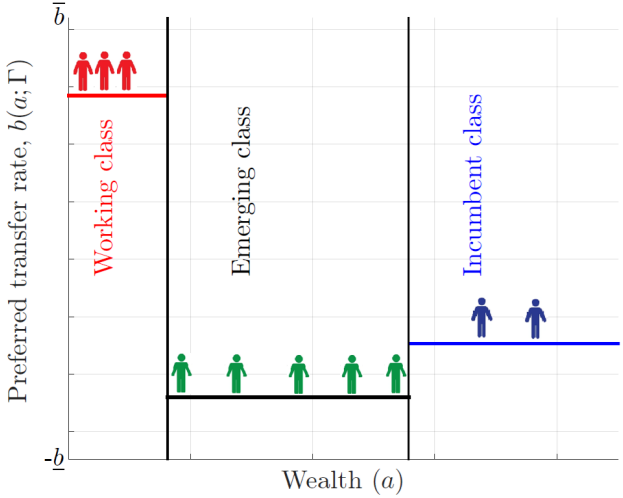
American Experience: Intuition

- $t = 0$: Wealthy and Unequal ⇒ Many Aspirational Voters (AV) ⇒ Low b



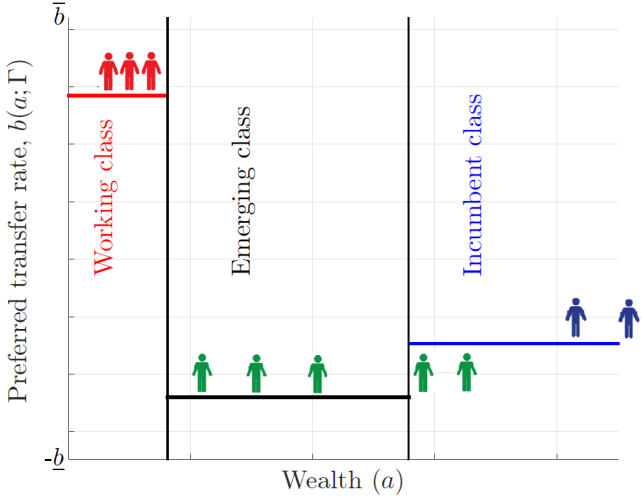
American Experience: Intuition

- $t = 0$: Low b $\Rightarrow$ Low τ $\Rightarrow$ Agents save



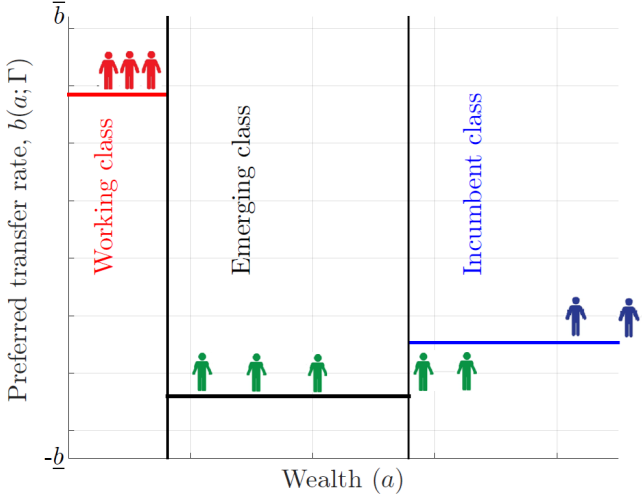
American Experience: Intuition

- $t = 0$: Low b $\Rightarrow$ Low τ $\Rightarrow$ Agents save $\Rightarrow$ Γ shifts right



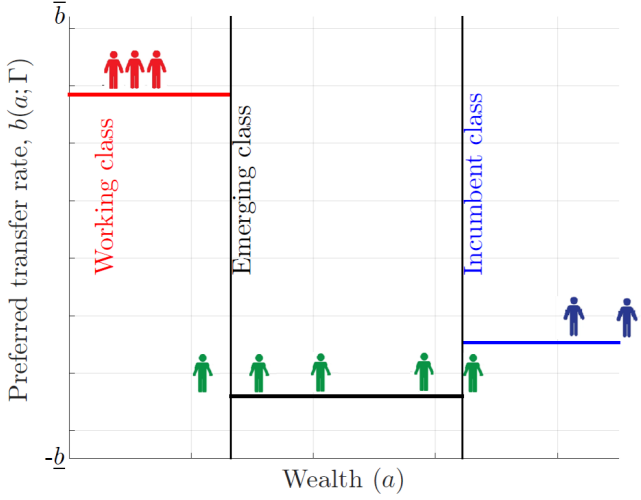
American Experience: Intuition

- $t = \Delta : \Gamma$ shifts right $\Rightarrow$ Everyone wealthier



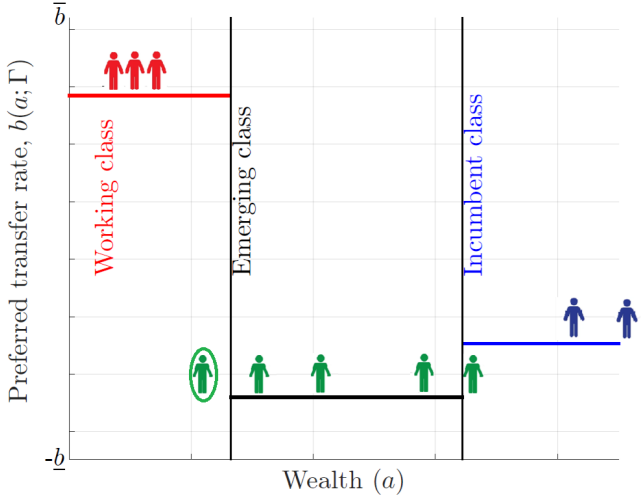
American Experience: Intuition

- $t = \Delta : \Gamma$ shifts right $\Rightarrow$ Everyone wealthier $\Rightarrow \uparrow a^o$



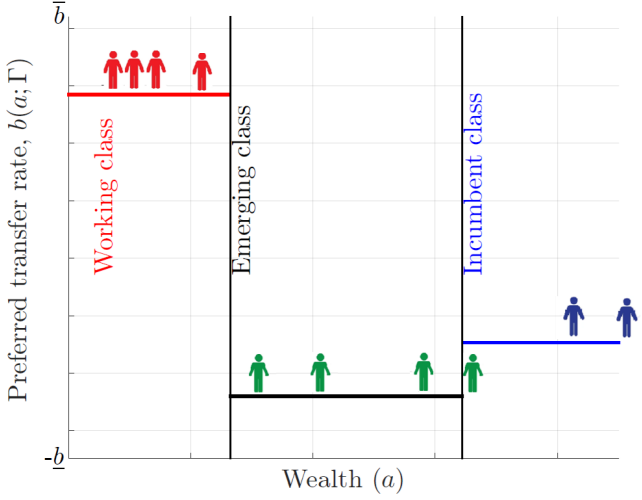
American Experience: Intuition

- $t = \Delta$: Poorest AV didn't save enough



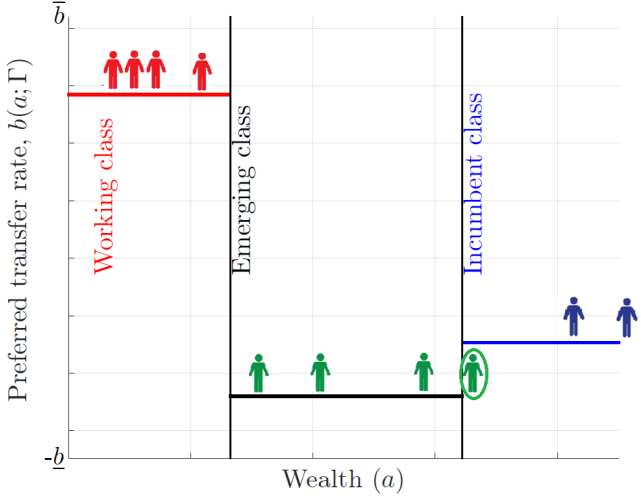
American Experience: Intuition

- $t = \Delta$: Poorest AV didn't save enough $\Rightarrow$ join the Working Class



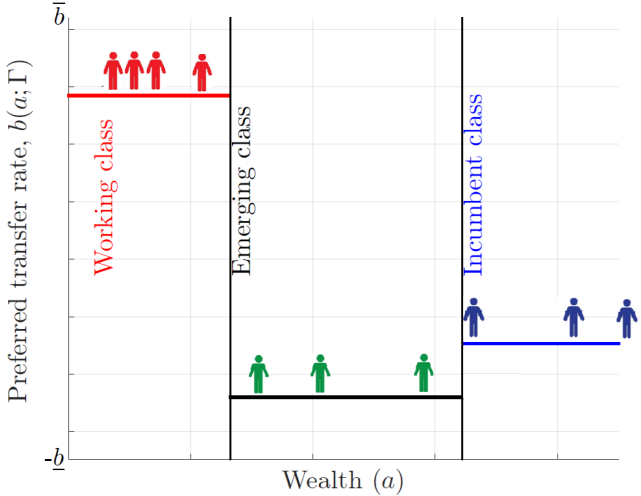
American Experience: Intuition

- $t = \Delta$: Wealthiest AV saved enough



American Experience: Intuition

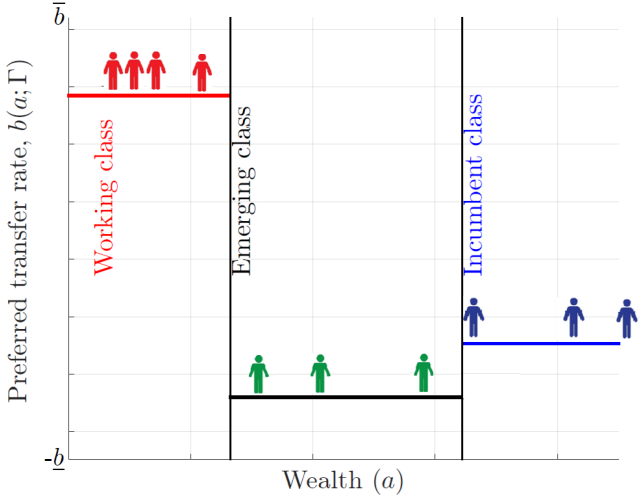
- $t = \Delta$: Wealthiest AV saved enough $\Rightarrow$ join the Incumbent Class



American Experience: Intuition

- $t = \Delta$: Mass of AV shrinks $\Rightarrow$ b goes up

Mathematical intuition



American Experience: Qualitative predictions

1. Increasing social benefits.
2. Shrinking of the middle class, expansion of the lower and upper classes.
3. Increasing wealth inequality.
4. Wealth accumulation.

- Theory consistent with the **fall of the American Dream**

Quantitative Exercise

Question Can the model systematically explain the evolution of the welfare state ?

Quantitative exercise

- **24** countries from all continents

Quantitative exercise

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- **Inputs**:

Quantitative exercise

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  - *Main moment:* social benefits at $t = 0$. Estimation method

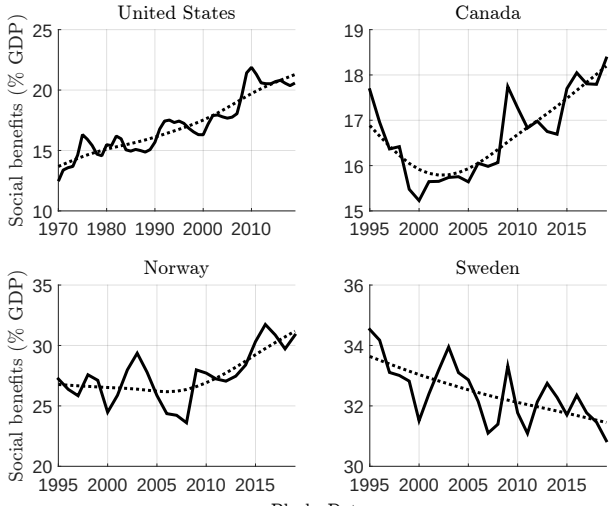
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- **Simulation:**

Quantitative exercise

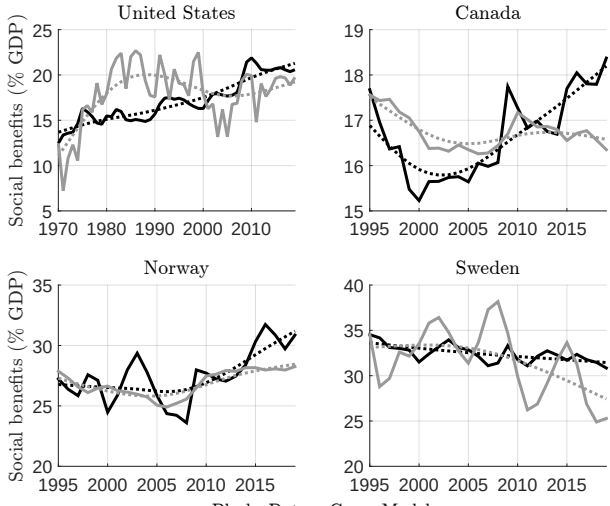
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  - *Main moment:* social benefits at $t = 0$. Estimation method
- **Simulation:**
  - Given Γ_0 and parameters, simulate future evolution of (b_t, Γ_t).

Countries in the Intro: Data versus Model

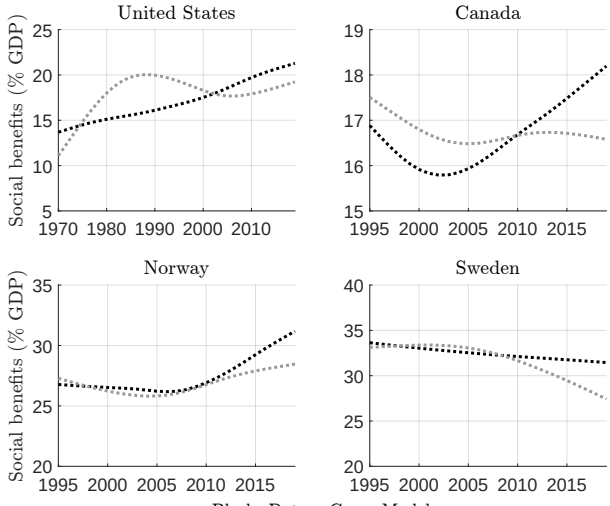


Black: Data

Countries in the Intro: Data versus Model



Countries in the Intro: Data versus Model



Extensions

Extensions

Theory

Extensions

Theory

1. Labor and capital tax $\surd$

Extensions

Theory

1. Labor and capital tax $\surd$
2. Transfers to entrepreneurs and workers $\surd$

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Quantitative Exercise

Extensions

Theory

1. Labor and capital tax $\surd$
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Quantitative Exercise

1. Simulations using only social benefits in cash $\surd$

Extensions

Theory

1. Labor and capital tax $\surd$
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1. Simulations using only social benefits in cash $\surd$
2. Counterfactual Analysis (Canada, USA, Sweden) $\surd$

Extensions

Theory

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   - Limited role of government changes in the trend of the Welfare State!

Extensions

Theory

1. Labor and capital tax $\surd$
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Quantitative Exercise

1. Simulations using only social benefits in cash $\surd$
2. Counterfactual Analysis (Canada, USA, Sweden) $\surd$
   - Limited role of government changes in the trend of the Welfare State!
3. *Future work:* Role of immigration (e.g. Canada and Sweden), aging population, ...

Conclusions

- Size of the Welfare State depends on Middle class of Aspirational voters
- Evolution of the Welfare State depends on Wealth & Inequality
- Theory predicts the trends of social benefits in 18 out 24 countries

Thanks!!!

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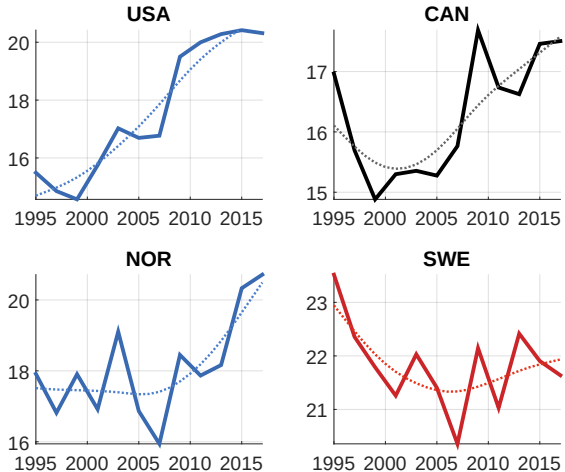
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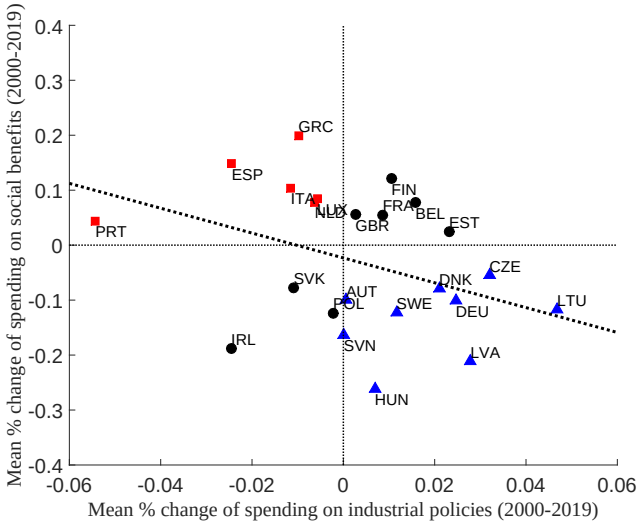
Supplementary Material

The Evolution of Net Social Benefits

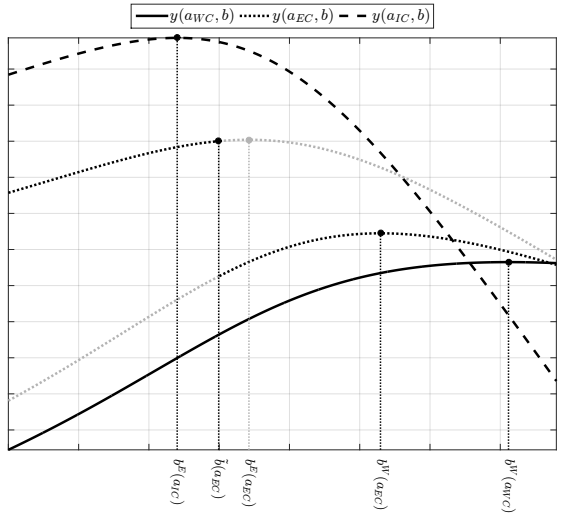


Main

Social Benefits versus Business Policies



Main


$y(a_{WC}, b)$
$y(a_{EC}, b)$
$y(a_{IC}, b)$
$\theta_E(a_{IC})$
$\tilde{b}(a_{EC})$
$\theta_E(a_{EC})$
$\theta_W(a_{EC})$
$\theta_W(a_{WC})$

Main

The Three Classes: Related Literature

- Emerging Class: Prospects of Upward Mobility Hypothesis
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- Incumbent Class: Interest group theories of financial development
  La Porta et al. (2000); Rajan and Zingales (2003); Rajan and Ramcharan (2011)

Main

Occupational Choice

Depends on two endogenous wealth thresholds:

1. Occupational threshold: $\tilde{a}(b_t, \Gamma_t)$
   ($\Pi_t \geq w_t \ell + T_t$)

2. Minimum collateral to get credit: $\hat{a}(b_t, \Gamma_t)$
   (Credit constraints á la Holmstrom and Tirole (1997))

- **Occupational choice** is made based on $a_t^O \equiv \max\{\tilde{a}_t, \hat{a}_t\}$.

The Three Classes: The Industrial Revolution in Britain



Main

The Three Classes: The Industrial Revolution in Britain



Main

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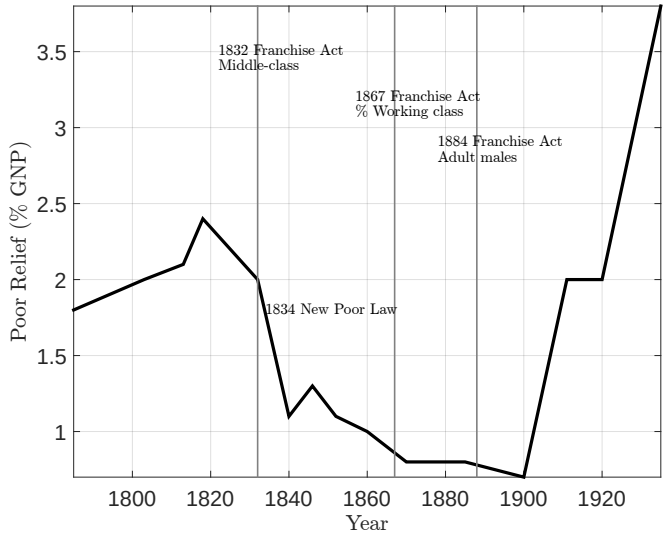
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The Three Classes: The Industrial Revolution in Britain



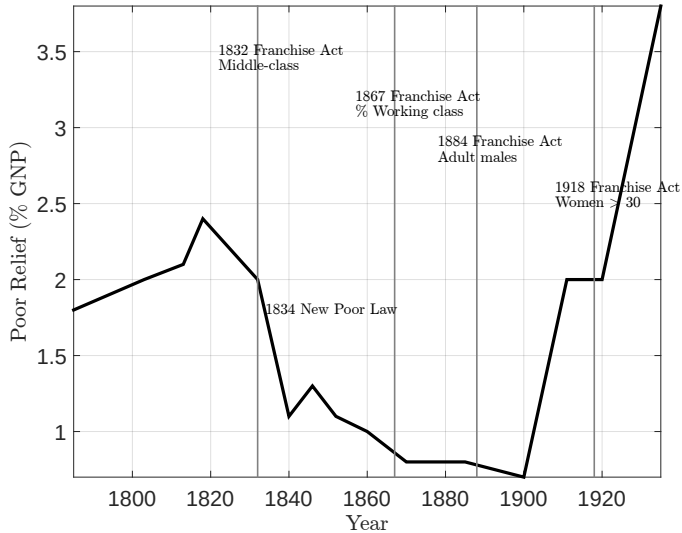
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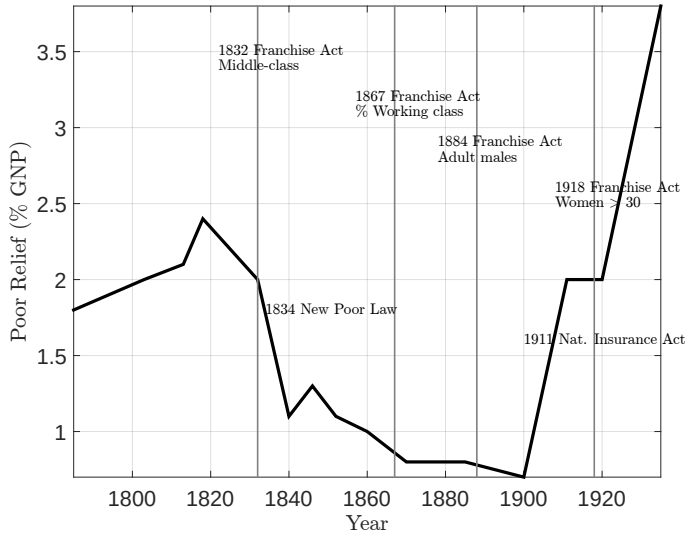
Main

The Three Classes: The Industrial Revolution in Britain



Main

The Three Classes: The Industrial Revolution in Britain



Main

Forward-looking government

- The government solves:

$$max_b\{\int v_t(a,b)d\Gamma_t(a)\}$$

- The PE condition is:

$$\int_{a<a^o(b,\Gamma_t)} \frac{(d_b w_t \ell + d_b T_t)}{y_t(a)} d\Gamma_t(a) + \int_{a\geq a^o(b,\Gamma_t)} \frac{d_b p_t}{y_t(a)} d\Gamma_t(a) = d_b \tau_t \int \frac{a}{y(a)} d\Gamma_t(a) + e^{\rho t}\left(\int_t^{+\infty} d_b \tau_s \frac{1}{r-\tau_s} e^{-\rho s} ds\right) + \frac{1}{\rho}$$

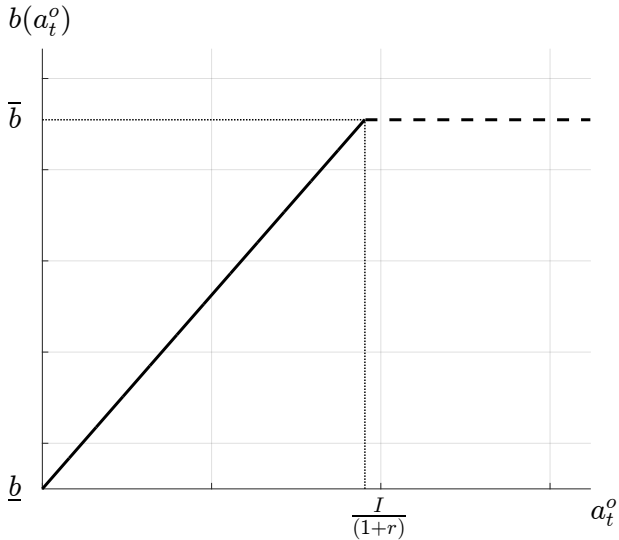
- **Observation** The evolution of b depends on Γ evaluated at each a

Main

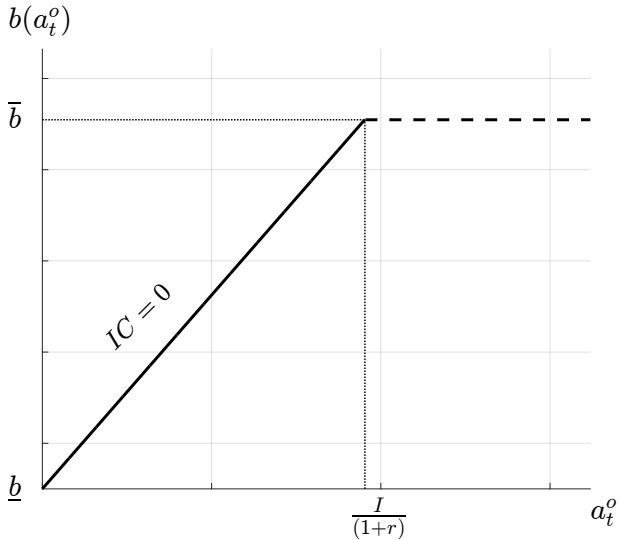
The **inequality** → **policy** link: $1 - \Gamma_t(a^o(b_t, \Gamma_t)) = e^*$

The inequality → policy link: $a_t^o = \Gamma_t^{-1}(1 - e^*)$

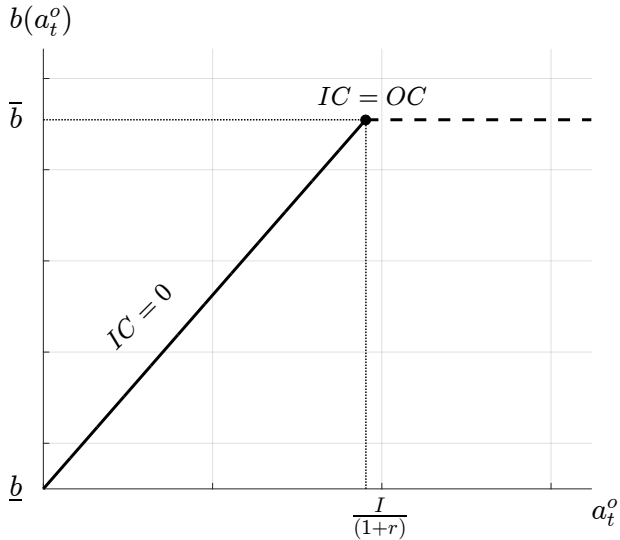
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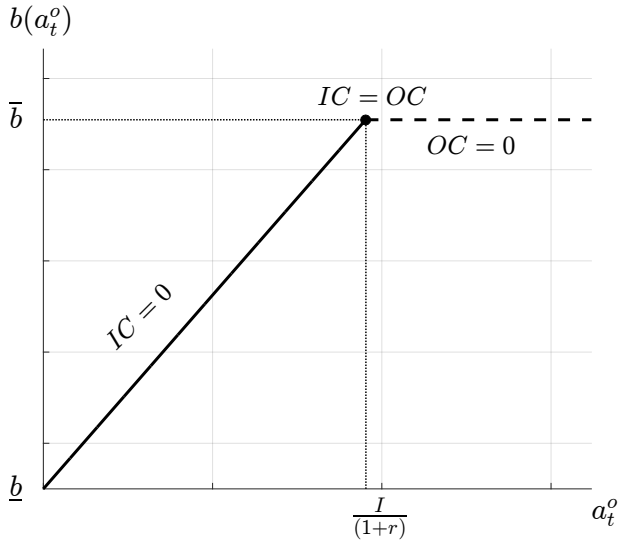
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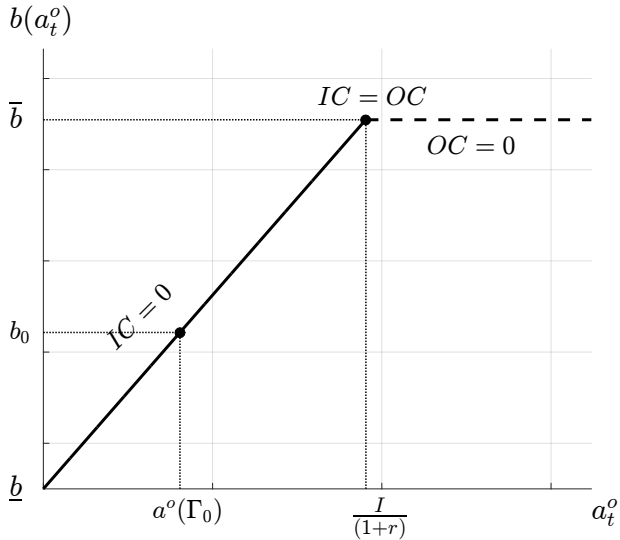
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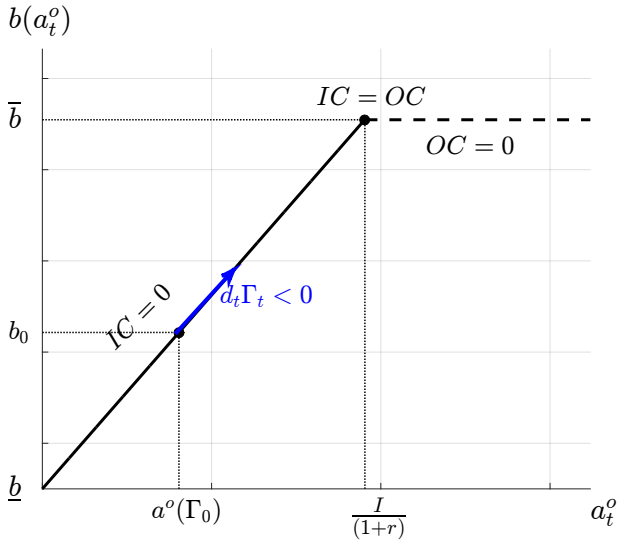
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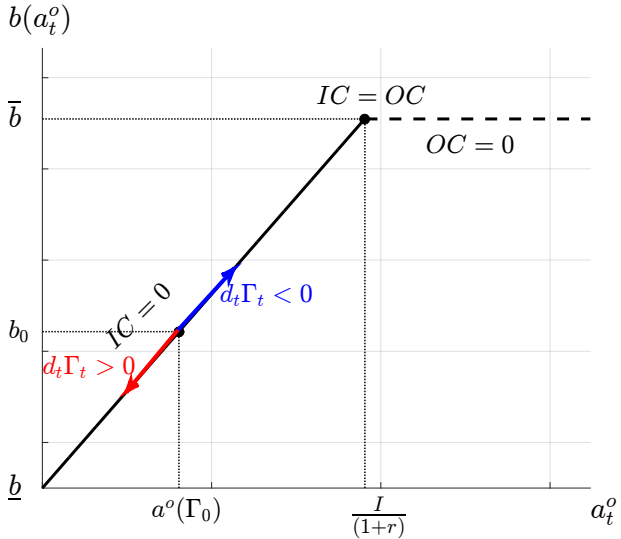


The inequality → policy link: $a_t^o = \Gamma_t^{-1}(1 - e^*)$



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Main



Stationary Equilibrium

Steady-state: $d_t \Gamma_t(a) = 0$

$$\tilde{H}(\Gamma^*, s = \theta^* \cdot y) = 0 \qquad (HJB) + (KFE)$$
$$\Rightarrow \theta^* = 0$$
$$\Rightarrow \boxed{\tau^* = r - \rho}$$

- **Result** *There is a unique stationary tax-rate*: τ^*

Stationary Equilibrium

Steady-state distribution (Γ^*)

$$r - \rho = \frac{b^* \Gamma^*(\hat{a}^*) \cdot y(\Gamma^*)}{A^*} \qquad (BB)$$

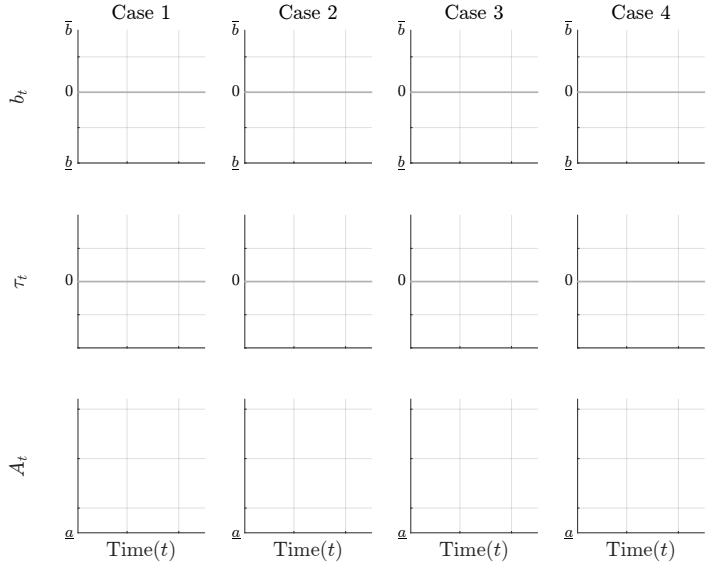
$$a^o* = \tilde{\psi}(\Gamma^*) \qquad (OC)$$

$$b^* = \tilde{\phi}(\Gamma^*) \qquad (PE)$$

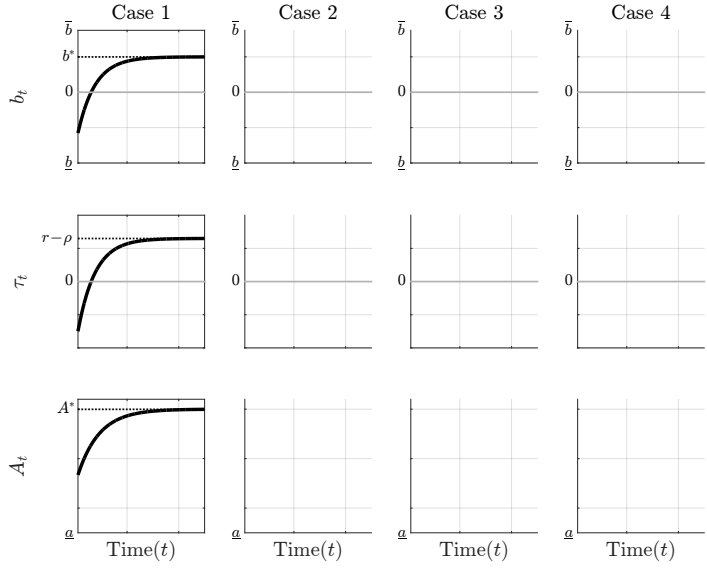
- **Result** Γ^* *is non-unique:* there is a set (A^*, Γ^*) that solves the system.
  - Similar result in the neoclassical model + politics.
    Krusell and Rios-Rull (1996, 1999)

Main

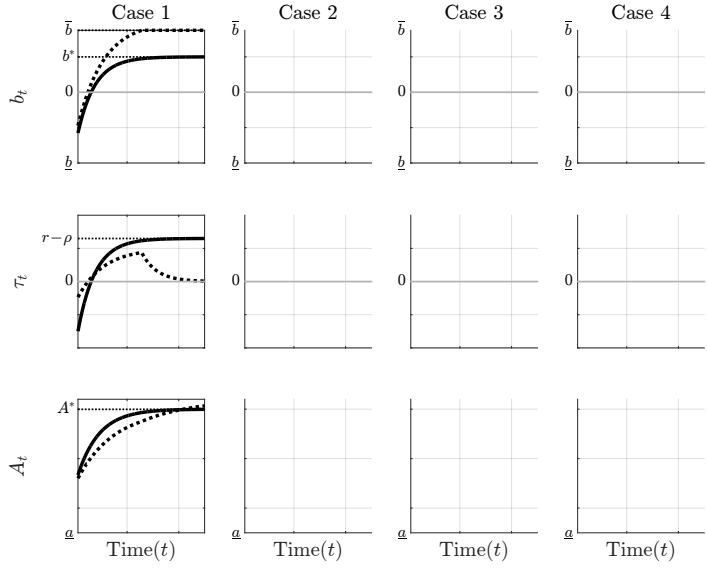
Cases are function of $r - \rho$ and Γ_0



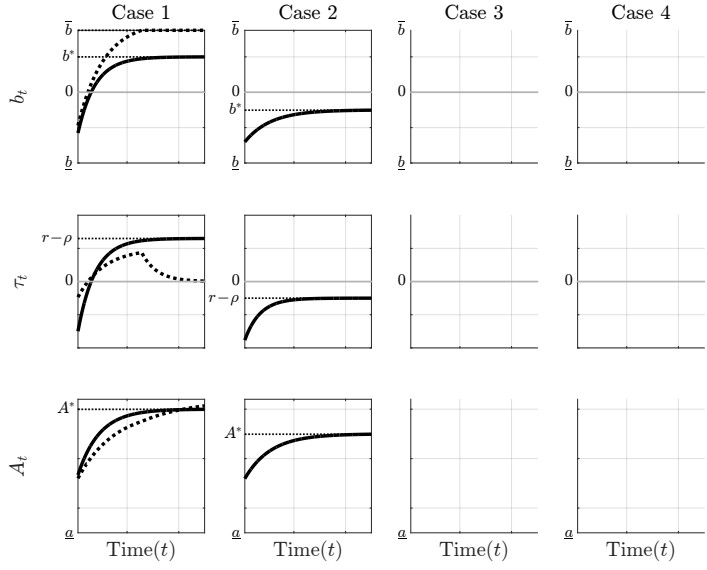
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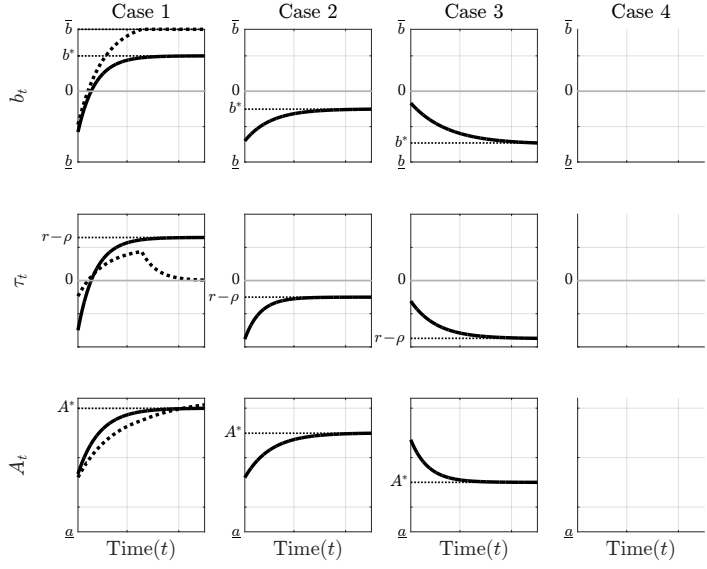
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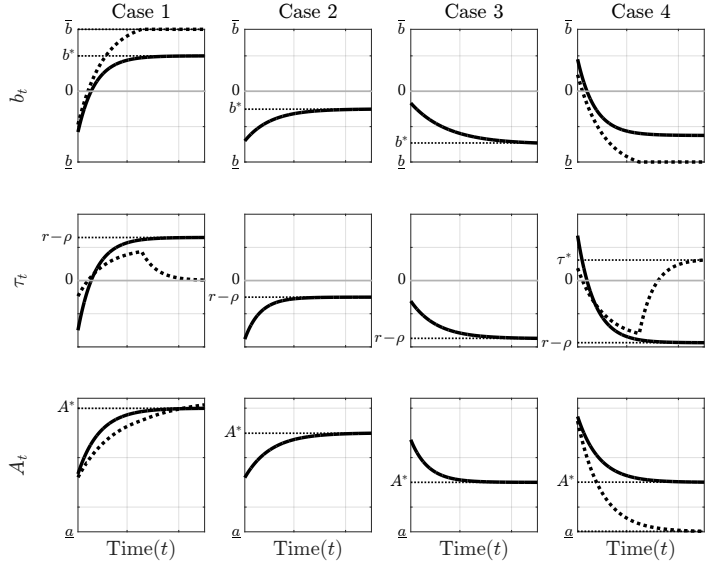
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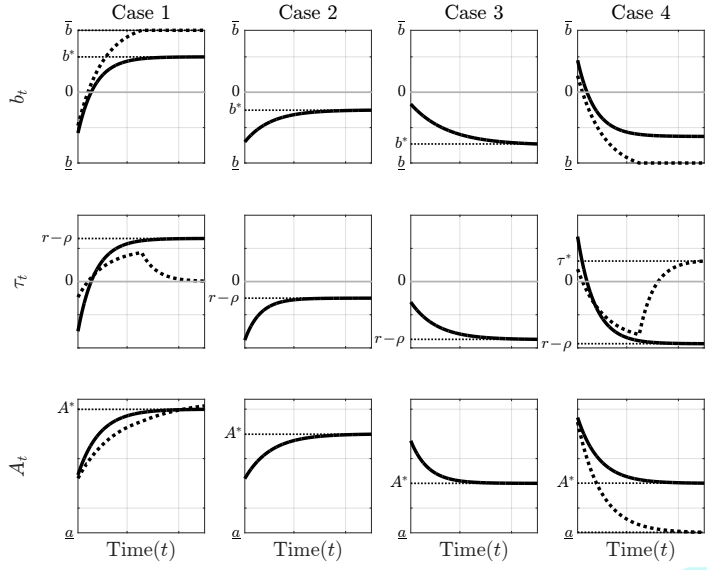
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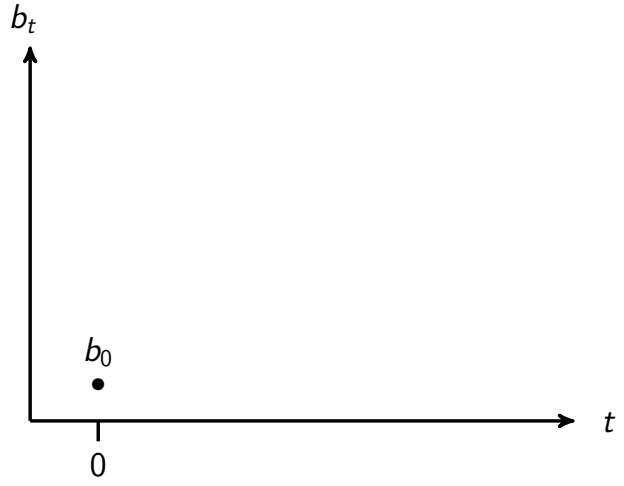


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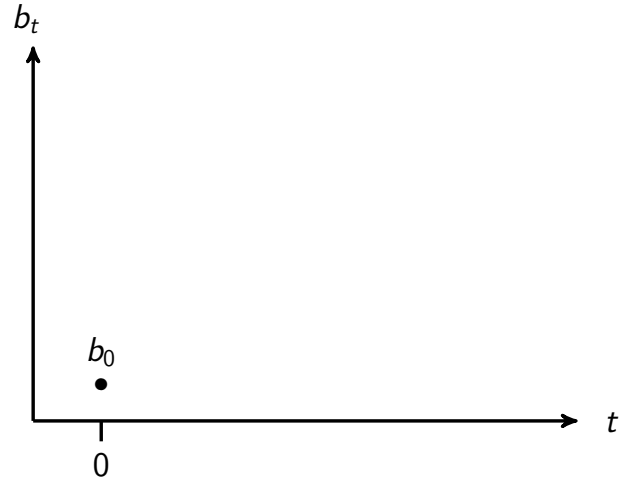


Details Main

Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



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- $\tau_0 < \tau^* = r - \rho$
  - $\theta_0 > 0 \Rightarrow$ agents save

Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



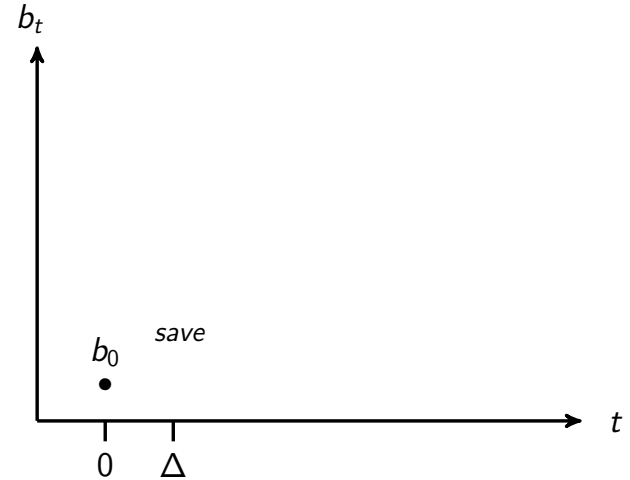
- $\tau_0 < \tau^* = r - \rho$
  - $\theta_0 > 0 \Rightarrow \Gamma_\Delta$ FOSD Γ_0 (Γ shifts right)

Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



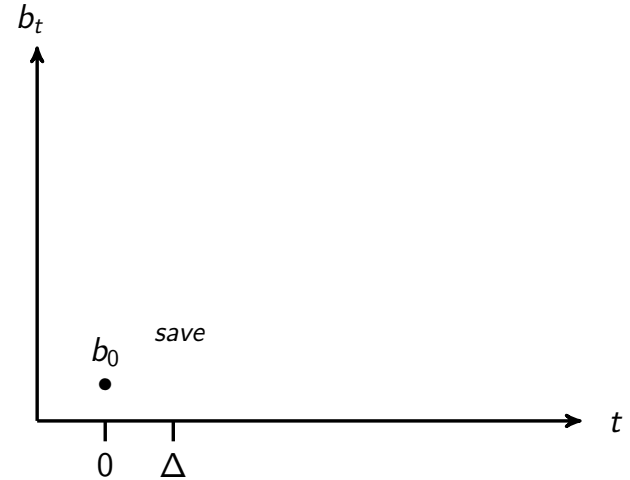
1. More entrepreneurs: $1 - \Gamma_\Delta(a^o(b_0, \Gamma_0)) > e^*$

Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



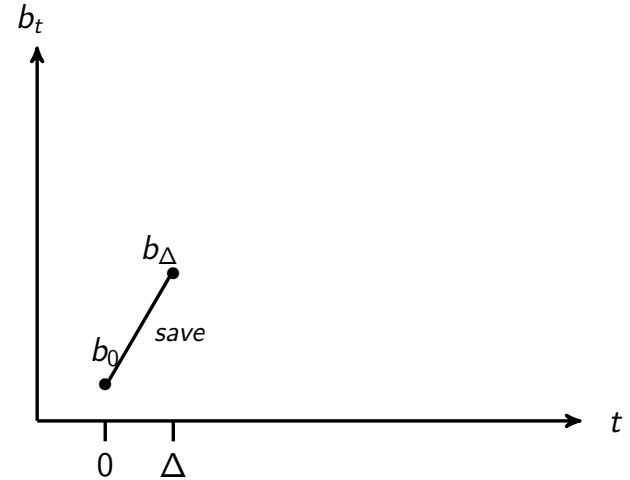
1. More entrepreneurs: $1 - \Gamma_\Delta(a^o(b_0, \Gamma_0)) > e^*$
2. More competition ($\downarrow \Pi$): $a^o(b_0, \Gamma_\Delta) > a^o(b_0, \Gamma_0)$

Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



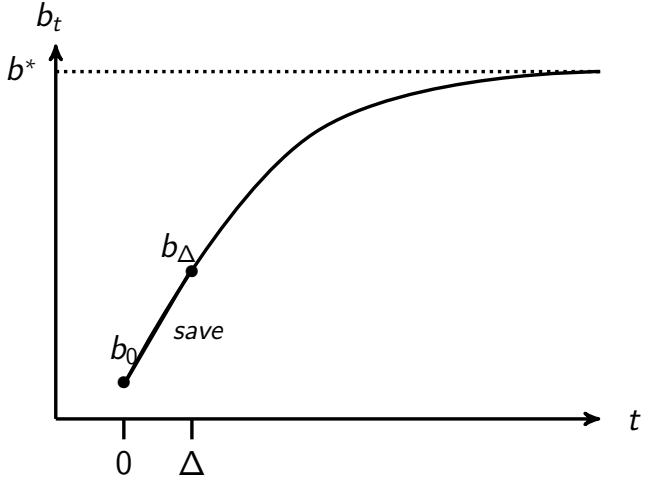
1. More entrepreneurs: $1 - \Gamma_\Delta(a^o(b_0, \boldsymbol{\Gamma_\Delta})) > e^*$ (net effect)
2. More competition ($\downarrow \Pi$): $a^o(b_0, \Gamma_\Delta) > a^o(b_0, \Gamma_0)$

Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



- Too many entrepreneurs: $1 - \Gamma_\Delta(a^o(b_0, \Gamma_\Delta)) > e^*$
  - Government: increases b to raise $a^o \Rightarrow \boldsymbol{b_\Delta > b_0}$

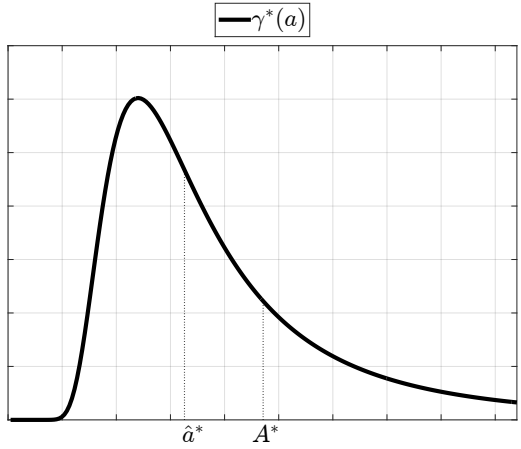
Transfer rate dynamics: Γ_0 such that $\tau_0 < r - \rho$



- b_t keeps increasing as long as $\theta_t > 0$
  - When $\theta_t = 0 \Rightarrow b_t = b^*$ Main

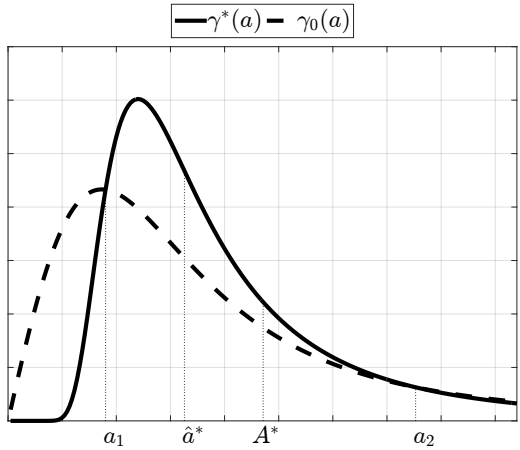
MPS in a Wealthy Country

- Capital unconstrained country ($A^* > \hat{a}^*$)



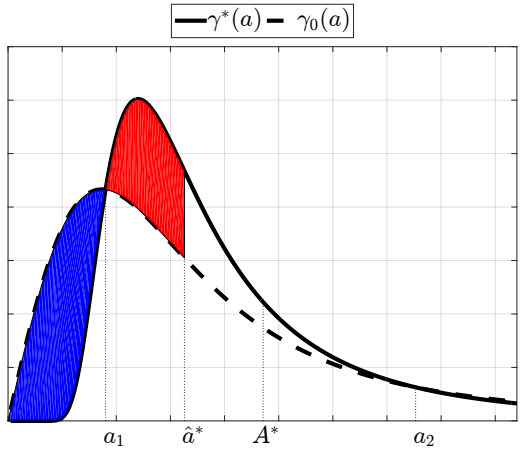
MPS in a Wealthy Country

- γ_0 more unequal than γ^* (*double-crossing*)



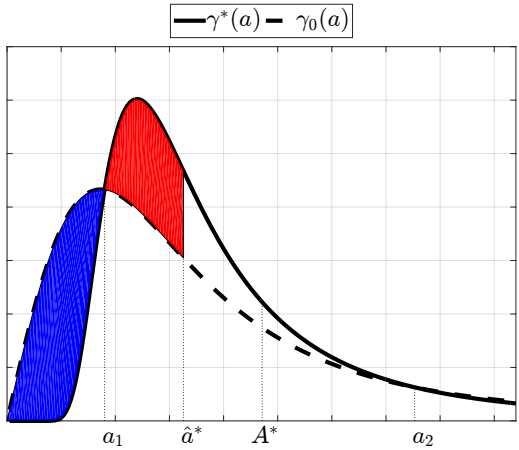
MPS in a Wealthy Country

- More unequal ⇒ Less entrepreneurs: $1 - \Gamma_0(\hat{a}^*) < 1 - \Gamma^*(\hat{a}^*)$



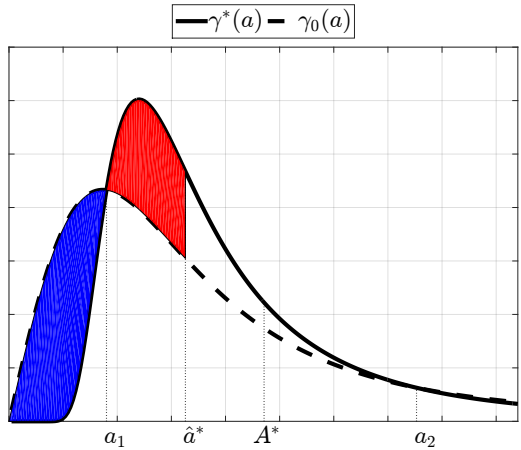
MPS in a Wealthy Country

- Net effect: $1 - \Gamma_0(\hat{a}_0) < 1 - \Gamma^*(\hat{a}^*) \Rightarrow \boldsymbol{b_0 < b^*} \Rightarrow \tau_0 < r - \rho$



MPS in a Wealthy Country

- $\tau_0 < r - \rho \Rightarrow$ *b* increasing over time



Estimation Method

- Set of parameters $\Psi = (r, \phi, I, R, \ell, \rho, \omega)_{1x7}$

Estimation Method

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- Set of moments:

$$m(\Psi|\Gamma_0) = \begin{bmatrix} \boldsymbol{b_0 - P(\Gamma_0, \Psi)} \\ K_0/L_0 - K/L(\Gamma_0, \Psi) \\ I_0/Y_0 - Inv(\Gamma_0, \Psi) \\ Giniy_0 - Giniy(\Gamma_0, \Psi) \\ b_0 - P(\Gamma_\Delta, \Psi) \\ \mathbb{E}[a|\Gamma_0] - \mathbb{E}[a|\Gamma_\Delta] \\ Var[a|\Gamma_0] - Var[a|\Gamma_\Delta] \\ Gini[a|\Gamma_0] - Gini[a|\Gamma_\Delta] \end{bmatrix}_{8x1}$$

Estimation Method

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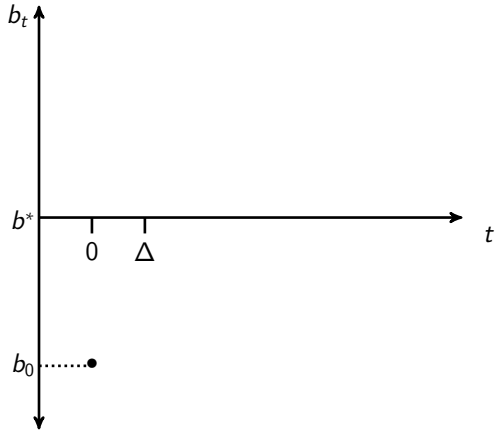
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- Solve: $\hat{\Psi} = argmin_\Psi \{ m(\Psi|\Gamma_0)' \; W \; m(\Psi|\Gamma_0) \}$

Main

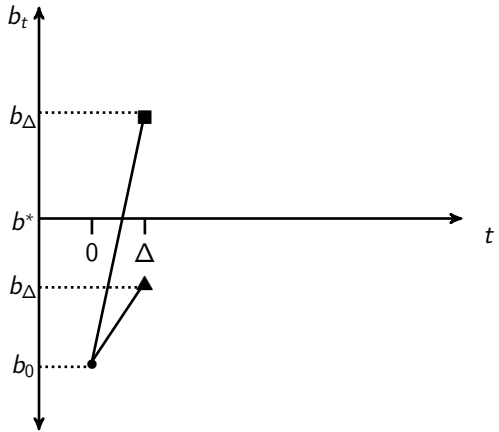
A permanent increase of productivity (MIT shock)

- At $t = 0$: $\uparrow Z \Rightarrow \uparrow e^* \Rightarrow 1 - G_0(\hat{a}(b^*)) < e^* \Rightarrow \downarrow \boldsymbol{b}$



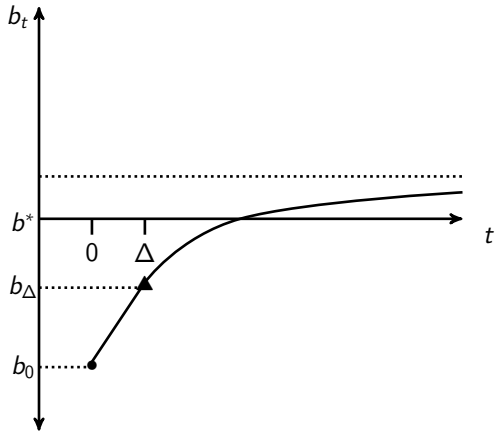
A permanent increase of productivity (MIT shock)

- At $t = \Delta$: $\boldsymbol{G}$ shifts right $\Rightarrow \uparrow \boldsymbol{b}$
  - $1 - G_\Delta(\hat{a}(b_0)) > e^*$



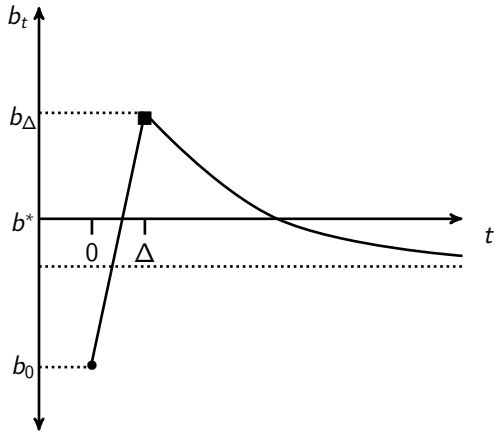
A permanent increase of productivity (MIT shock)

Case 1



A permanent increase of productivity (MIT shock)

Case 2



Main

The "Oscillatory" Behavior of τ

- *Example:* Suppose that $\uparrow b_t$ and $\uparrow A_t$. Recall:

$$\tau_t = \frac{b_t}{A_t} \cdot (1 - e^*) \cdot y(e = e^*)$$

The "Oscillatory" Behavior of τ

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  - τ may oscillate over time $\Rightarrow$ b may hit the *PC* before $\tau_t \to \tau^*$

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     - τ may oscillate over time $\Rightarrow$ b may hit the *PC* before $\tau_t \to \tau^*$
- The dynamics of b can still be characterized!

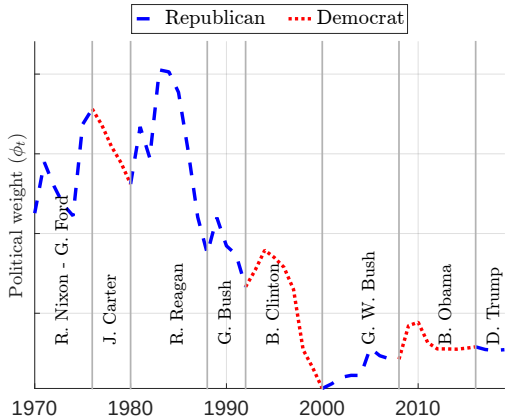
Counterfactual Analysis

Question Role of Politics in the Evolution of the Welfare State?

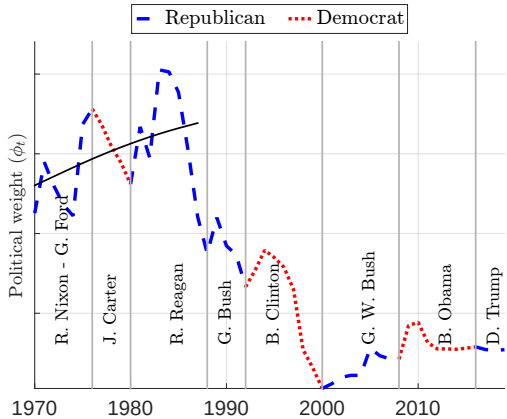
Counterfactual Analysis for the US

1. Find the sequence of Political Weights $\{\phi_t\}_{1970}^{2019}$ that matches $\{b_t\}_{1970}^{2019}$
2. Simulate the model for "extreme" alternative paths around $\{\phi_t\}_{1970}^{2019}$
3. **Question** Does the trend of social benefits change?

USA:The Evolution of the Political Weight

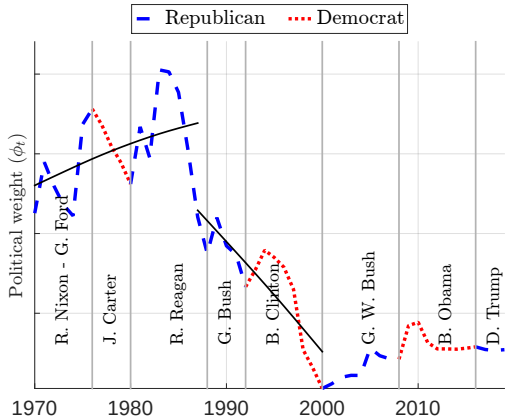


USA:The Evolution of the Political Weight



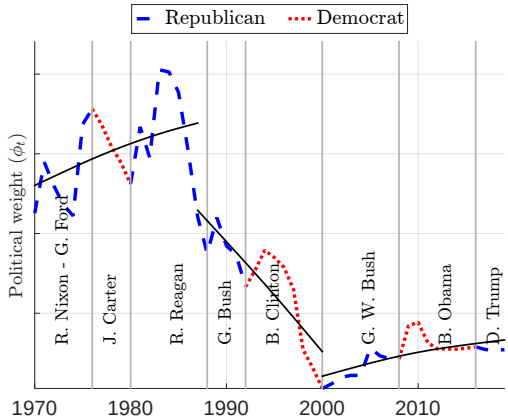
- 1970-1990: Pro-business trend ($\uparrow \phi$)

USA:The Evolution of the Political Weight



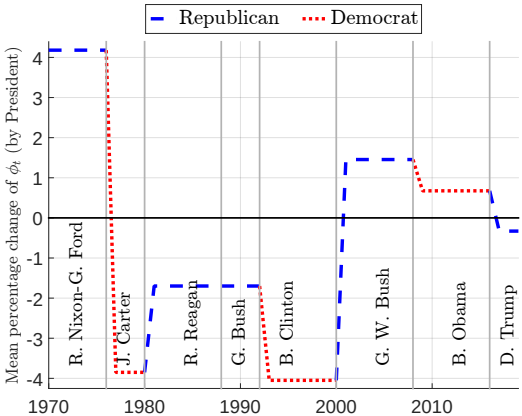
- 1990-2000: Pro-worker trend ($\downarrow \phi$)

USA:The Evolution of the Political Weight



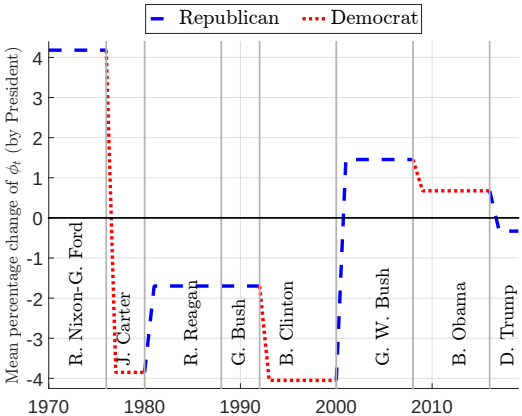
- 2000-present: moderate Pro-business trend ($\nearrow \phi$)

USA:The Evolution of the Political Weight



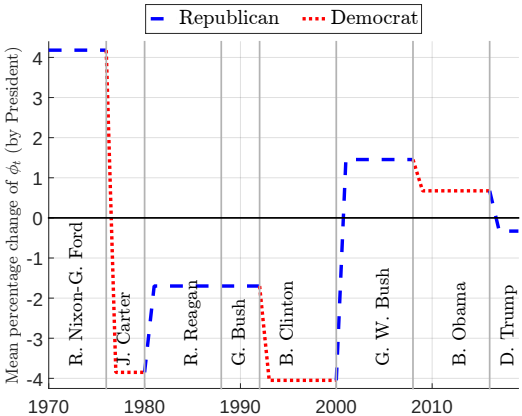
- **Republicans:** largest increases of ϕ

USA:The Evolution of the Political Weight



- **Democrats:** largest decreases of ϕ

USA:The Evolution of the Political Weight



- Behavior of ϕ consistent with partisan political perspectives

Question What would have been the evolution of b if?

1. Pro-worker scenario (Low ϕ): $\phi_t \times$ largest % drop

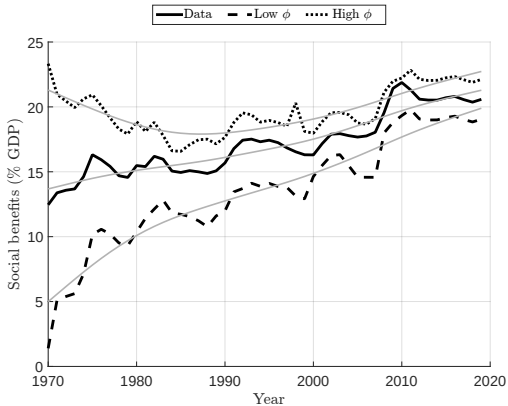
Question What would have been the evolution of b if?

1. Pro-worker scenario (Low ϕ): $\phi_t \times$ largest % drop
2. Pro-business scenario (High ϕ): $\phi_t \times$ largest % increase

Question What would have been the evolution of b if?

1. Pro-worker scenario (Low ϕ): $\phi_t \times$ largest % drop

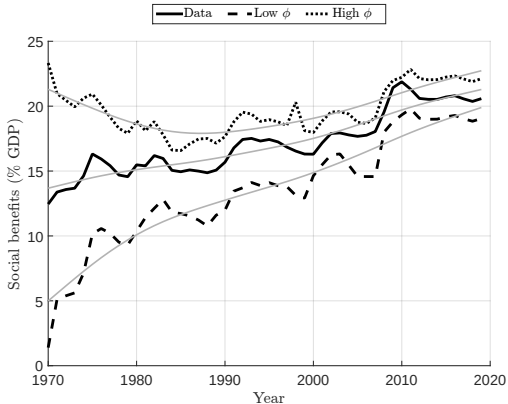
2. Pro-business scenario (High ϕ): $\phi_t \times$ largest % increase



Question What would have been the evolution of b if?

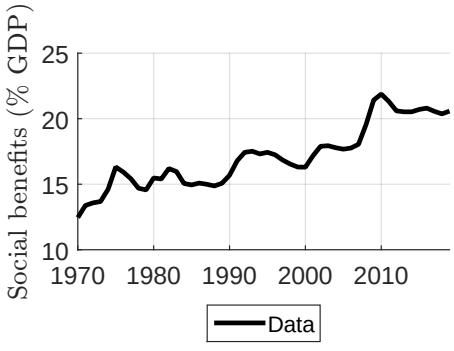
Main

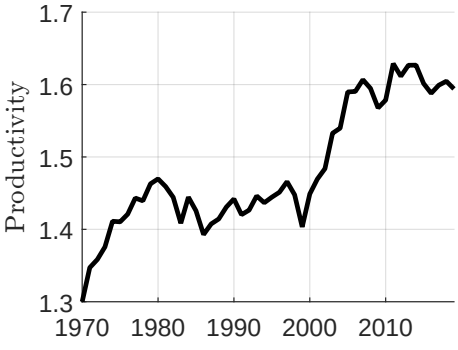
- Trend of b would have remained positive since 1990
- **Main message:** Limited role of politics in the evolution of the welfare state



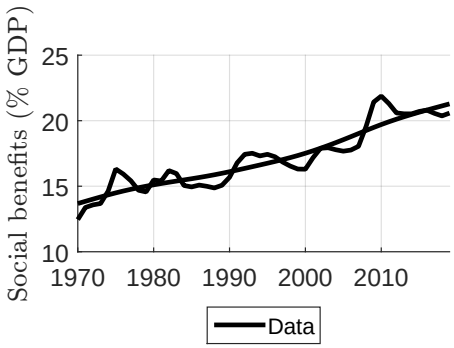
The Role of Productivity

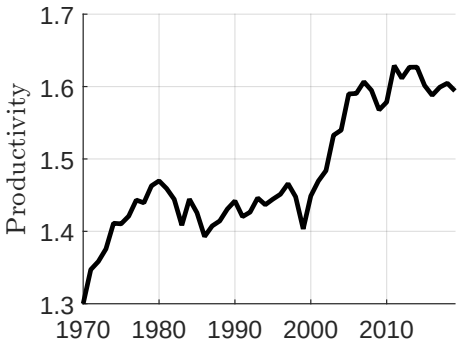
United States: Social benefits and Productivity



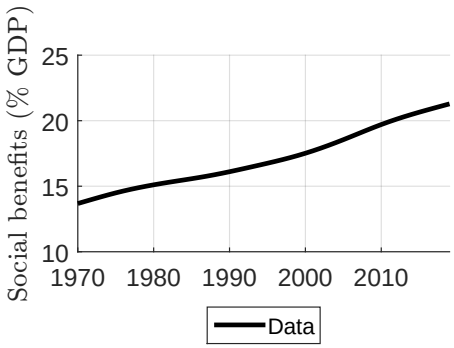


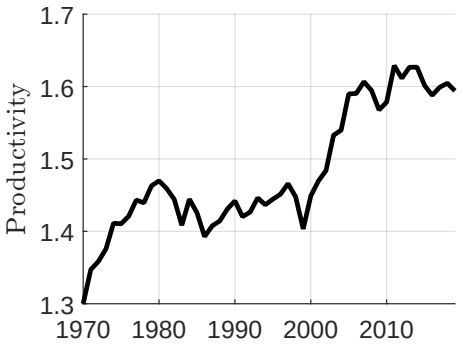
United States: Social benefits and Productivity



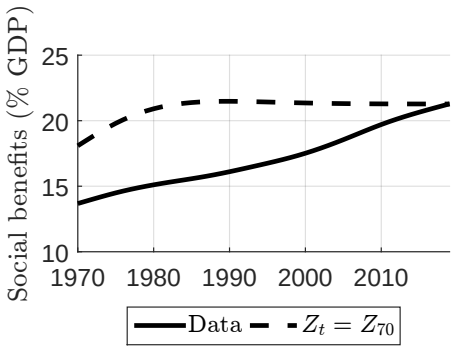


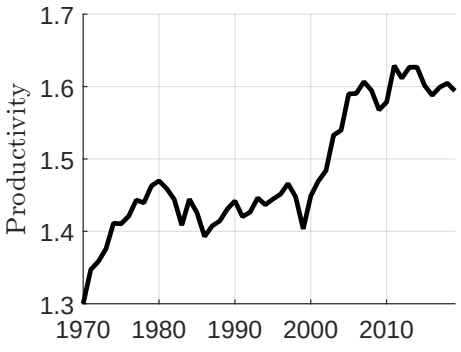
United States: Social benefits and Productivity



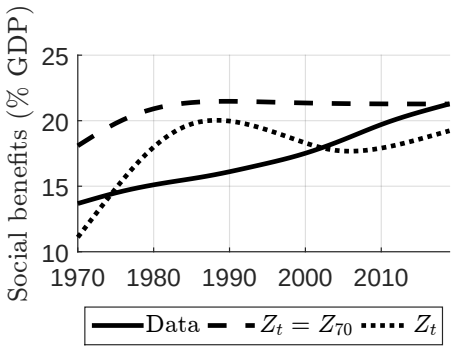


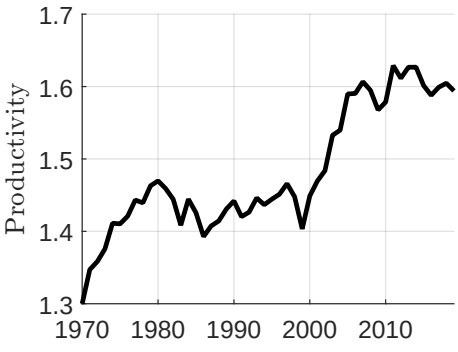
United States: Social benefits and Productivity





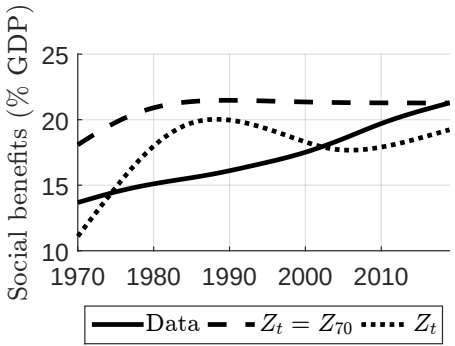
United States: Social benefits and Productivity

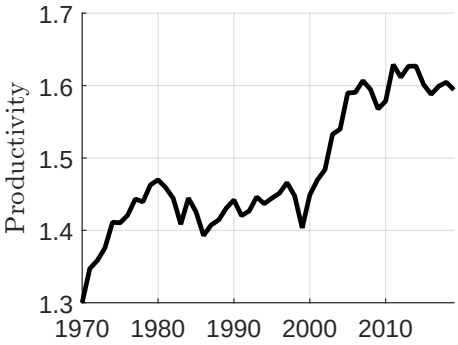




United States: Social benefits and Productivity

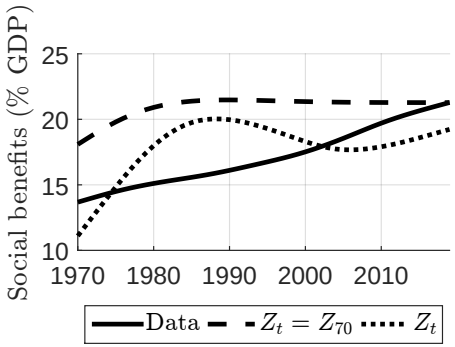
- Effects of increasing productivity?

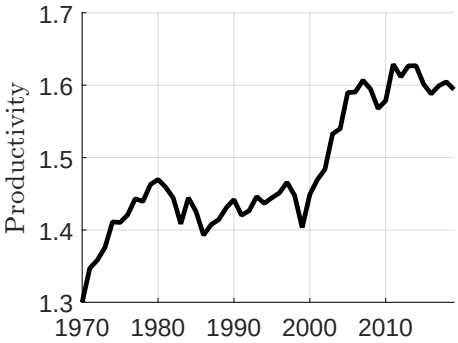




United States: Social benefits and Productivity

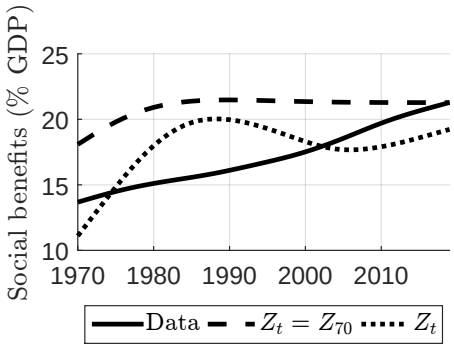
- 1st order effect: $\uparrow Z \Rightarrow \uparrow \Pi$ and $\downarrow a^o$

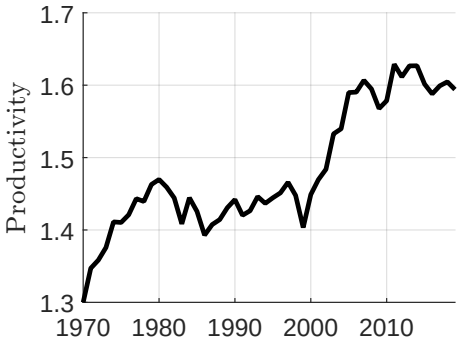




United States: Social benefits and Productivity

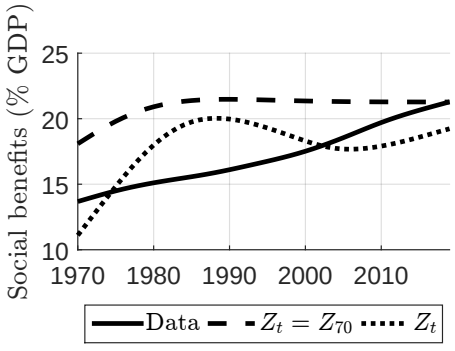
- 1st order effect: $\uparrow Z \Rightarrow \uparrow \Pi$ and $\downarrow a^o \Rightarrow \uparrow$ AV $\Rightarrow \downarrow b$

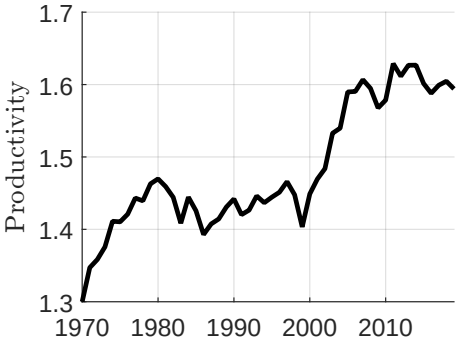




United States: Social benefits and Productivity

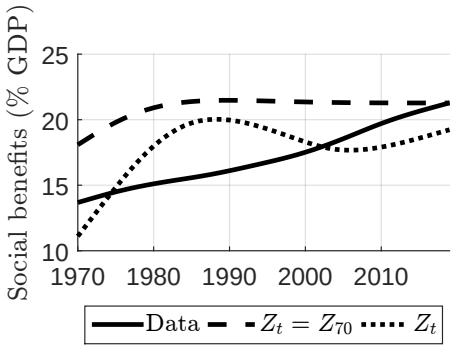
- Why social benefits going up despite increasing productivity?

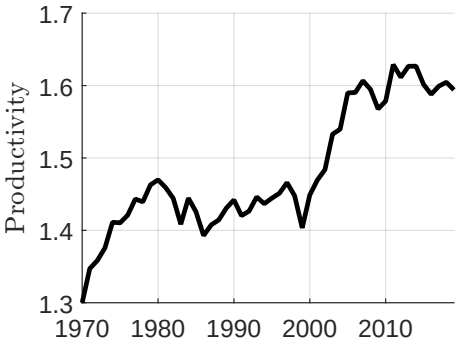




United States: Social benefits and Productivity

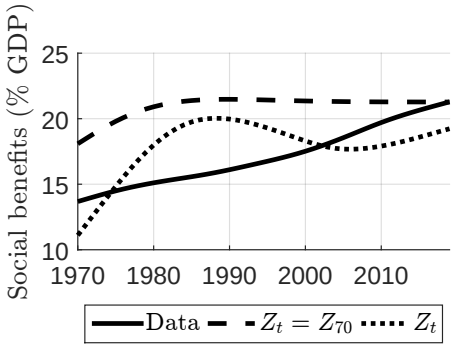
- 2nd order effect: $\downarrow b \Rightarrow \downarrow \tau \Rightarrow$ Agents save

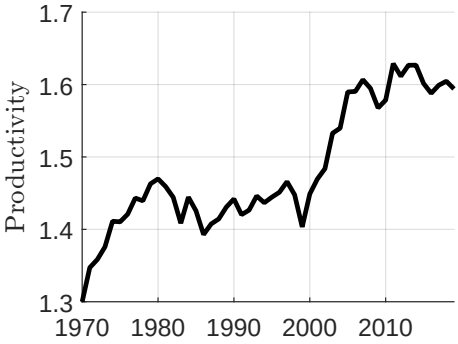




United States: Social benefits and Productivity

- 2nd order effect: $\downarrow b \Rightarrow \downarrow \tau \Rightarrow$ Agents save $\Rightarrow \downarrow$ mass of AV $\Rightarrow \uparrow b$





United States: Social benefits and Productivity

- 2nd order effect has dominated in the US! Main

